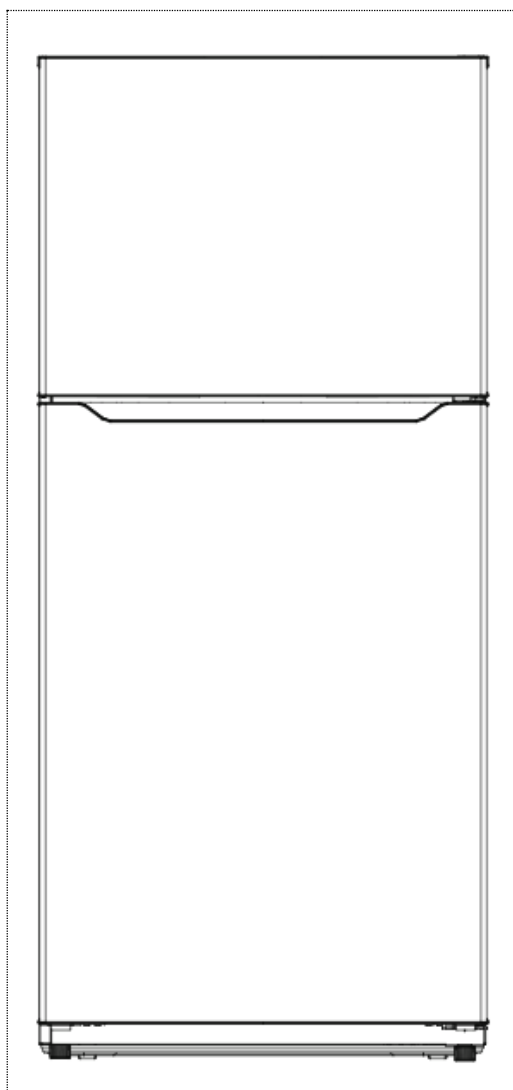


Service Manual

TMF No Frost Series

Applicable Models	Product Models	Customer model
HD-668FWEN	UR-BCD512WE-SQ	NS-RTM18SS2 NS-RTM18WH2



The picture in this service manual is only for reference, and specific appearance and configuration are subject to the real product.

This manual mainly teaches the method, the specific work skill needs engineer to accumulate through the daily work.

 WARNING

Important Safety Notice

There are special components used in this equipment which are important for safety. These parts are marked by  in the Schematic Diagrams, Circuit Board Diagrams, Exploded Views and Replacement Parts List. It is essential that these critical parts should be replaced with manufacturer's specified parts to prevent shock, fire or other hazards. Do not modify the original design without permission of manufacturer.

 WARNING

Important Safety Notice

The Maintenance Manual is only for the use of maintenance personnel with certain experience and background in electrical, electronic and mechanical field.

Any attempt to repair main devices may lead to personal injury and property loss.

Manufacturers or distributors are not responsible for the content of the Manual and interpretation thereof.

Midea Refrigerators

Technical Maintenance Manual

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1. Significant update notes (None)

2. Safety Warning

2.1 Warning for operation safety

Important Safety Instructions

	CAUTION RISK OF ELECTRIC SHOCK DO NOT OPEN	
	This symbol indicates that dangerous voltage constituting a risk of electric shock is present within your freezer.	
	This symbol indicates that there are important operating and maintenance instructions in the literature accompanying your freezer.	

WARNING

- 1) Read these instructions.
- 2) Keep these instructions.
- 3) Heed all warnings.
- 4) Follow all instructions.
- 5) Do not use this appliance near water.
- 6) Clean only with a damp cloth.
- 7) Do not block any ventilation openings.
- 8) Install in accordance with the manufacturer's instructions.
- 9) Do not install near any heat sources, such as radiators, heat registers, stoves, or other apparatus that produce heat.
- 10) Do not defeat the safety purpose of the polarized or grounding-type plug. A polarized plug has two blades with one wider than the other. A grounding type plug has two blades and a third grounding prong. The wide blade or the third prong are provided for your safety. If the provided plug does not fit into your outlet, consult an electrician for replacement of the obsolete outlet.
- 11) Protect the power cord from being walked on or pinched, particularly at plugs, convenience receptacles, and the point where they exit from the appliance.
- 12) Do not attempt to modify or extend the power cord of this appliance.
- 13) Unplug this appliance during lightning storms or when it will not be used for long periods of time.
- 14) Make sure that the available AC power matches the voltage requirements of this appliance.

CONNECTING ELECTRICITY

**WARNING****Electrical Shock Hazard.**

Plug into a grounded 3-prong outlet.
Do not remove the ground prong.
Do not use an adapter.

Failure to follow these instructions can result in death, fire, or electrical shock.

**WARNING****Electric Shock Hazard**

Failure to follow these instructions can result in electric shock, fire, or death.

- 1) **WARNING**—Keep ventilation openings, in both the freezer and the built-in structure, clear of obstruction.
- 2) **WARNING**—Do not touch the interior of the freezer with wet hands. This could result in frost bite.
- 3) **WARNING**—Do not use mechanical devices or other means to accelerate the defrosting process, other than those recommended by the manufacturer.
- 4) **WARNING**—Do not damage the refrigerant circuit.
- 5) **WARNING**—Do not damage the refrigerant tubing when handling, moving, or using the freezer.
- 6) **WARNING—DANGER**—Never allow children to play with, operate, or crawl inside the freezer. Risk of child entrapment. Before you throw away your old freezer:
 - 6-1) Take off the doors
 - 6-2) Leave the shelves in place so that children may not easily climb inside
- 7) Unplug the freezer before carrying out user maintenance on it.
- 8) This freezer can be used by children age eight years and older and persons with reduced physical or mental capabilities or lack of experience and knowledge if they are given supervision or instruction concerning the use of the freezer in a safe way and understand the hazards involved. Children should not play with the freezer. Cleaning and maintenance should not be performed by children without supervision.
- 9) If a component part is damaged, it must be replaced by the manufacturer, its service agent, or similar qualified persons in order to avoid a hazard.
- 10) Please dispose of the freezer according to local regulations as the freezer contains flammable gas and refrigerant.
- 11) Follow local regulations regarding disposal of the freezer due to flammable refrigerant and gas. All refrigeration products contain refrigerants, which under the guidelines of federal law must be removed before disposal. It is the consumer's responsibility to comply with federal and local regulations when disposing of this product.
- 12) This freezer is intended to be used in household and similar environments.

13) Do not store or use gasoline or any flammable liquids inside or in the vicinity of this freezer.

14) Do not use extension cords or ungrounded (two-prong) adapters with this freezer. If the power cord is too short, have a qualified electrician install an outlet near the freezer. Use of an extension cord can negatively affect the freezer's performance.

Grounding requirement

This freezer must be grounded. This freezer is equipped with a cord having a grounding wire with a grounding plug. The plug must be inserted into an outlet that is properly installed and grounded.

Improper use of the grounding plug can result in a risk of electric shock. Consult a qualified electrician or service person if the grounding instructions are not completely understood, or if doubt exists as to whether the freezer is properly grounded.

2.2 Safety instruction for refrigerant

⚠ WARNING



Explosion Hazard.

Keep flammable materials and vapors, such as gasoline, away from freezer. Failure to do so can result in fire, explosion, or death.



Safety instruction for refrigerant

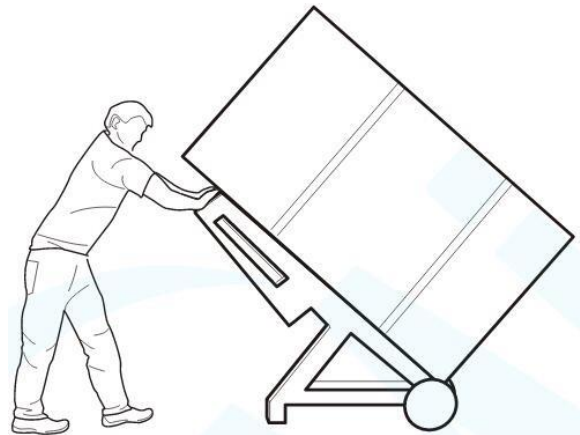
DANGER—Risk of Fire or Explosion. Flammable Refrigerant Used. To Be Repaired Only By Trained Service Personnel. Do Not Use Mechanical Devices. Do Not Puncture Refrigerant Tubing. CAUTION—Risk of Fire or Explosion. Flammable Refrigerant Used. Consult Repair Manual/Owner's Guide Before Attempting To Service This Product. All Safety Precautions Must be Followed. CAUTION—Risk of Fire or Explosion. Dispose of Properly In Accordance With Federal Or Local Regulations. Flammable Refrigerant Used. CAUTION—Risk of Fire or Explosion Due To Puncture Of Refrigerant Tubing; Follow Handling Instructions Carefully. Flammable Refrigerant Used.

3. Installation and commissioning

3.1 Handling

Handling

- 1) Protect the refrigerator in moving it, Same as shown as left photo, please move it by handcart with cushion
- 2) Remove all packing materials and bottom cushion, the move into house for placement
- 3) After moving it to appropriate location, wait for 2 hours before power on.



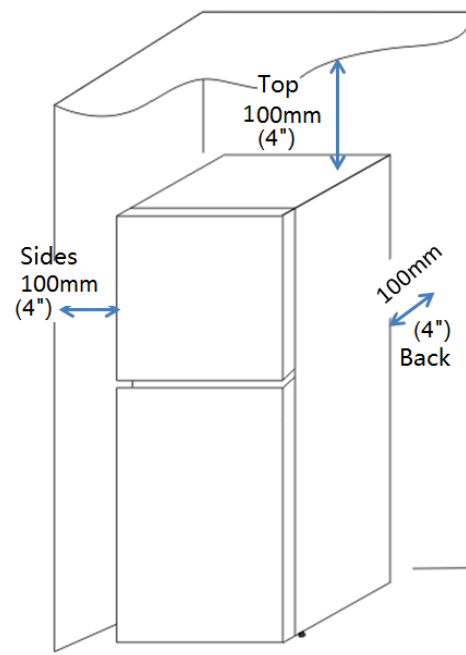
3.2 Door Disassembly and Assembly

When the whole refrigerator cannot enter the room, the door can be disassembled, then assembled after entering separately.

3.3 Installation location

Installation location

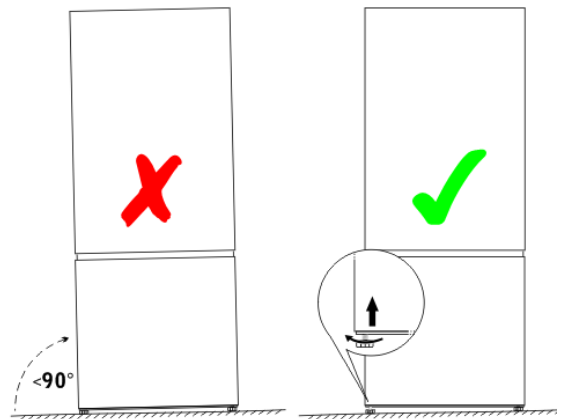
Please select a ventilated place to place the refrigerator, and reserve space according to the recommended size in the picture, which is conducive to heat dissipation, performance improvement and energy consumption reduction.



3.4 Leveling of the refrigerator

Leveling of the refrigerator

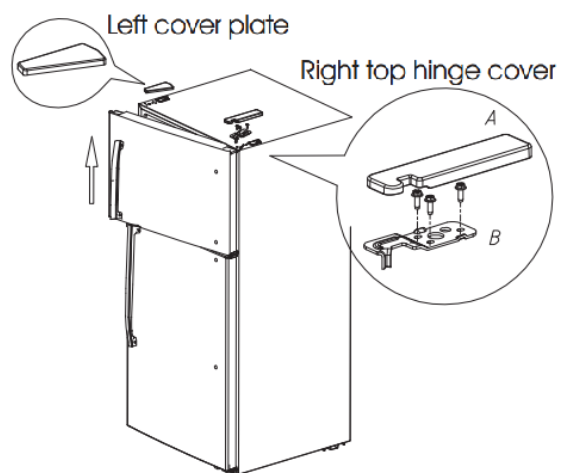
If the refrigerator cannot be placed steadily, adjust the footing to level it. Turn the feet clockwise to raise the refrigerator; turn the feet counterclockwise to lower the refrigerator.



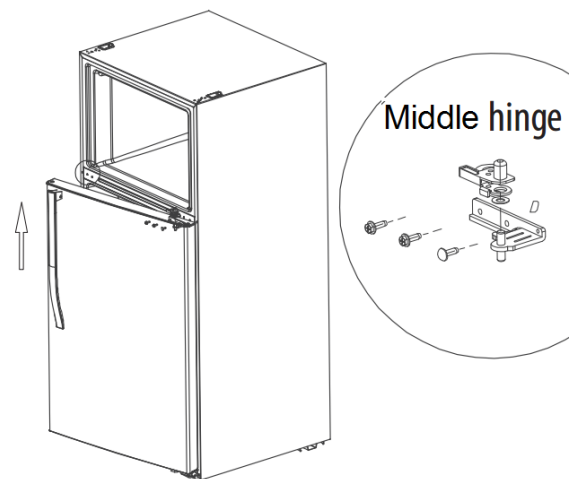
3.5 Left or right open door reversal

Door reversal

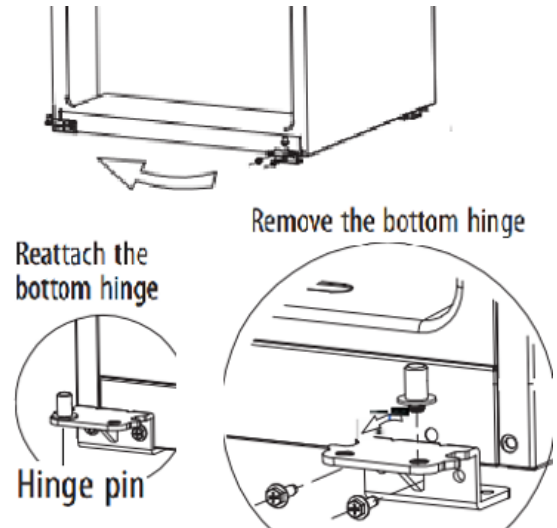
- 1) Power off and remove all food from the fridge.
- 2) Dismantle the upper hinge cover, screws and upper hinge from the right top, and remove the cover plate from the left top.



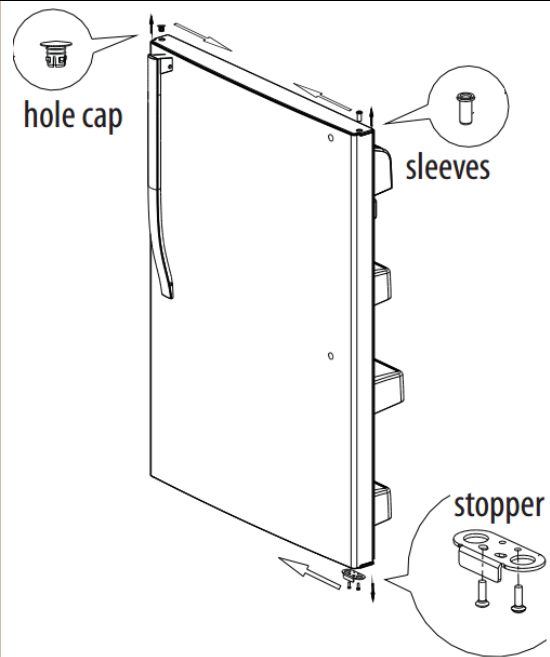
- 3) Take off the freezer door.
- 4) Unscrew the three screws holding the center hinge, remove the hinge, then lift the refrigerator door up and away from bottom hinge.



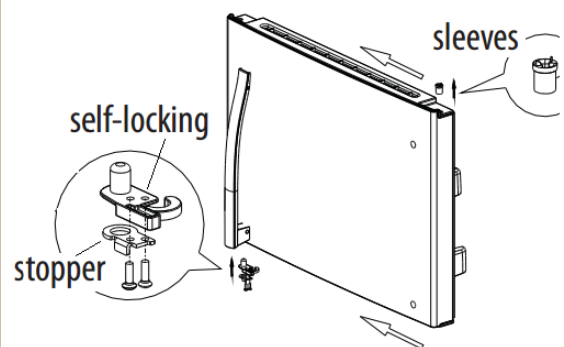
- 5) Remove the bottom hinge and screws.
 6) Unscrew the hinge shaft from the right side of the bottom hinge and move it to the left side, then secure the bottom hinge to the bottom left side of the refrigerator.



- 7) Dismantle the sleeves, hole cap, screws and door stopper on the refrigerator door, and assemble them on the other side.

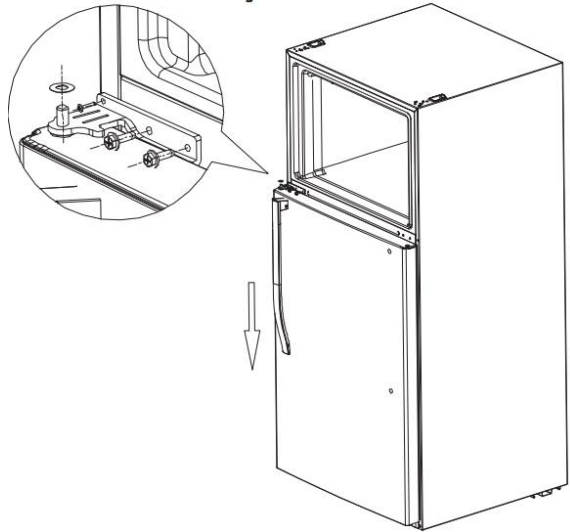


- 8) Dismantle the sleeves, and assemble them on the other side.
 9) Remove the freezer door self-locking block and stopper, save the to reuse. Take out the left self-locking block and stopper from the accessory plastic bag in the refrigerator and installed on the left of door.

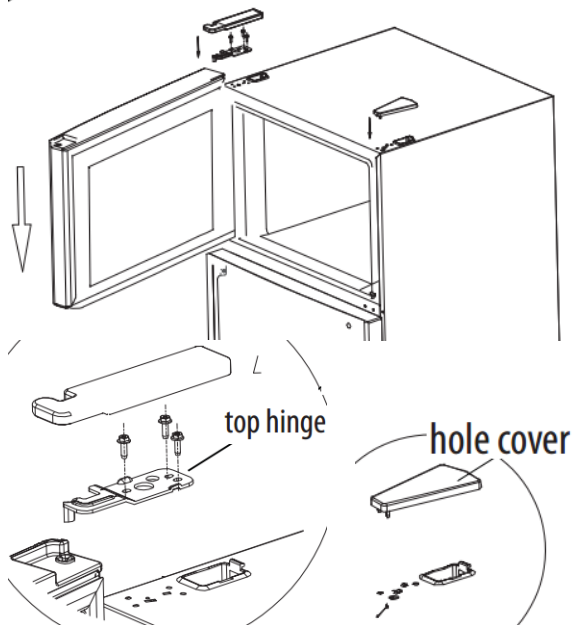


10) Put down the refrigerator door on the bottom hinge, take out the middle hinge of the other side from the accessory plastic bag, then assemble the middle hinge.

Attach the left center hinge



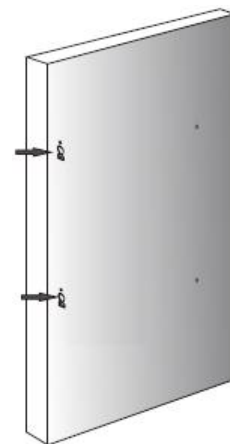
11) Put down the freezer door on the middle hinge, then take out the top hinge and cover of other side from the accessory plastic bag. Then assemble the left top hinge, hinge cover and hole cover.



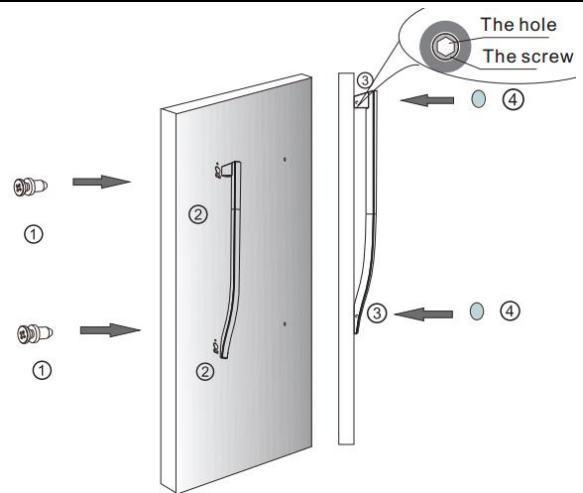
3.6 Installation of handle (Option)

Installation of handle

- 1) Remove the hole covers on the front of the door.
- 2) Take out the door handle, bolts and screw covers from the accessory plastic bag.



- 3) Fix the bolts① in the screw hole on the front of the door.
- 4) Install the door handle② onto the bolts.
- 5) Tighten the Allen screws③ in the door handle hole.
- 6) Install the screw cover④ on the screw of the door handle.

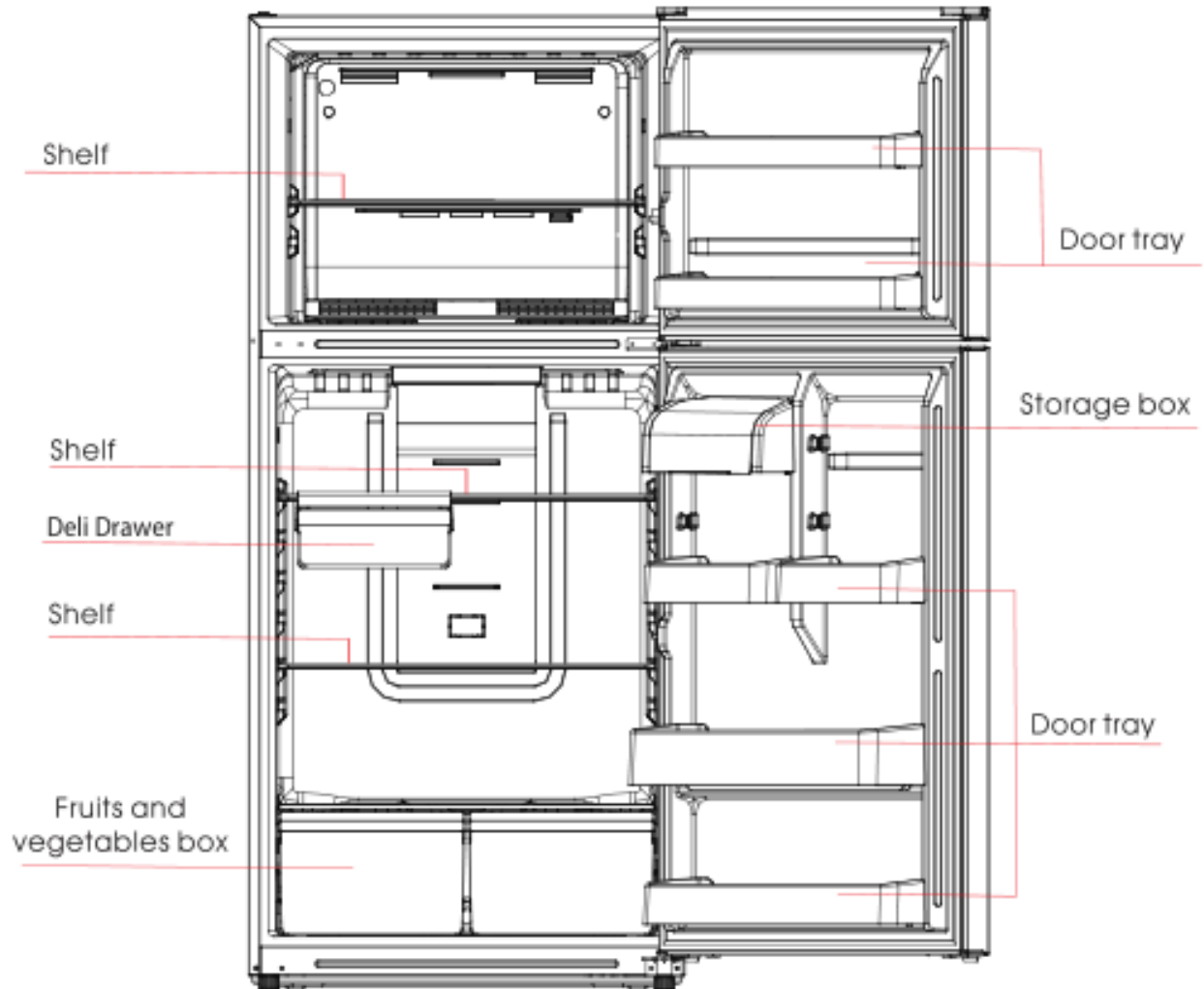


4. Main parts and external dimension

4.1 Main parts

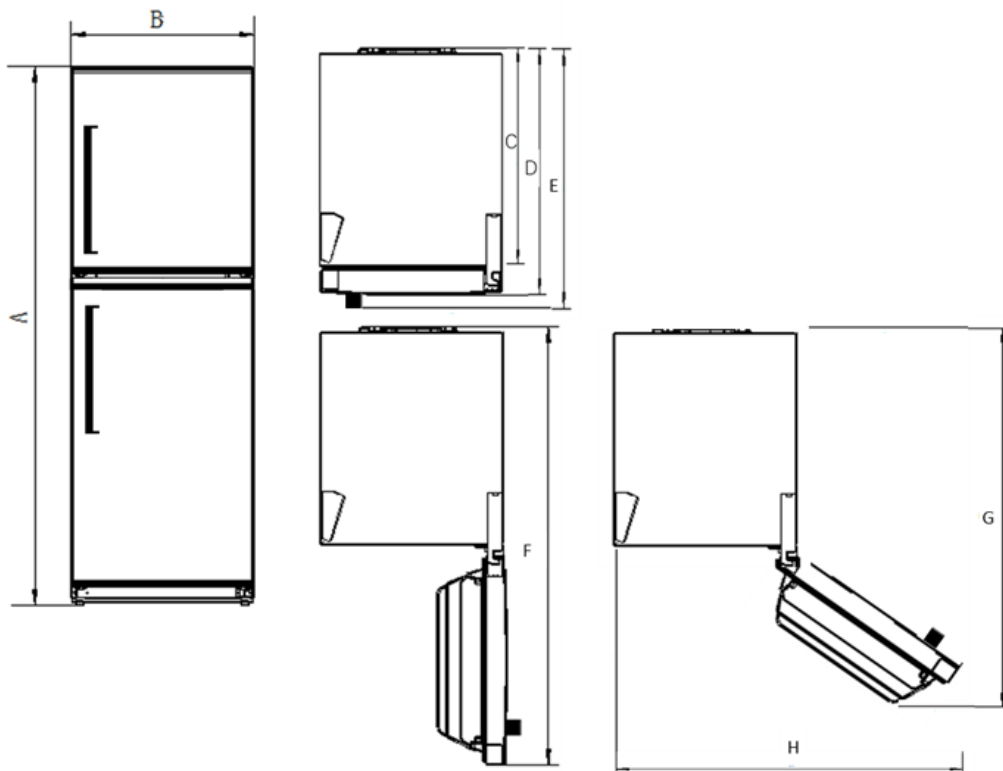
(The picture is only for reference, and specific appearance and configuration are subject to the real product)

(Freezing chamber)



4.2 External dimension

Description	Code	Size (mm)
Height to top of Cabinet	A	1692
Width	B	755
Depth w/Cabinet	C	680
Depth w/Door	D	780
Depth w/Handle	E	\
Depth (Door open 90 deg. w)	F	1495.5
Depth (Door open 135 deg. w)	G	1342
Width (Door open 135 deg. w)	H	1275



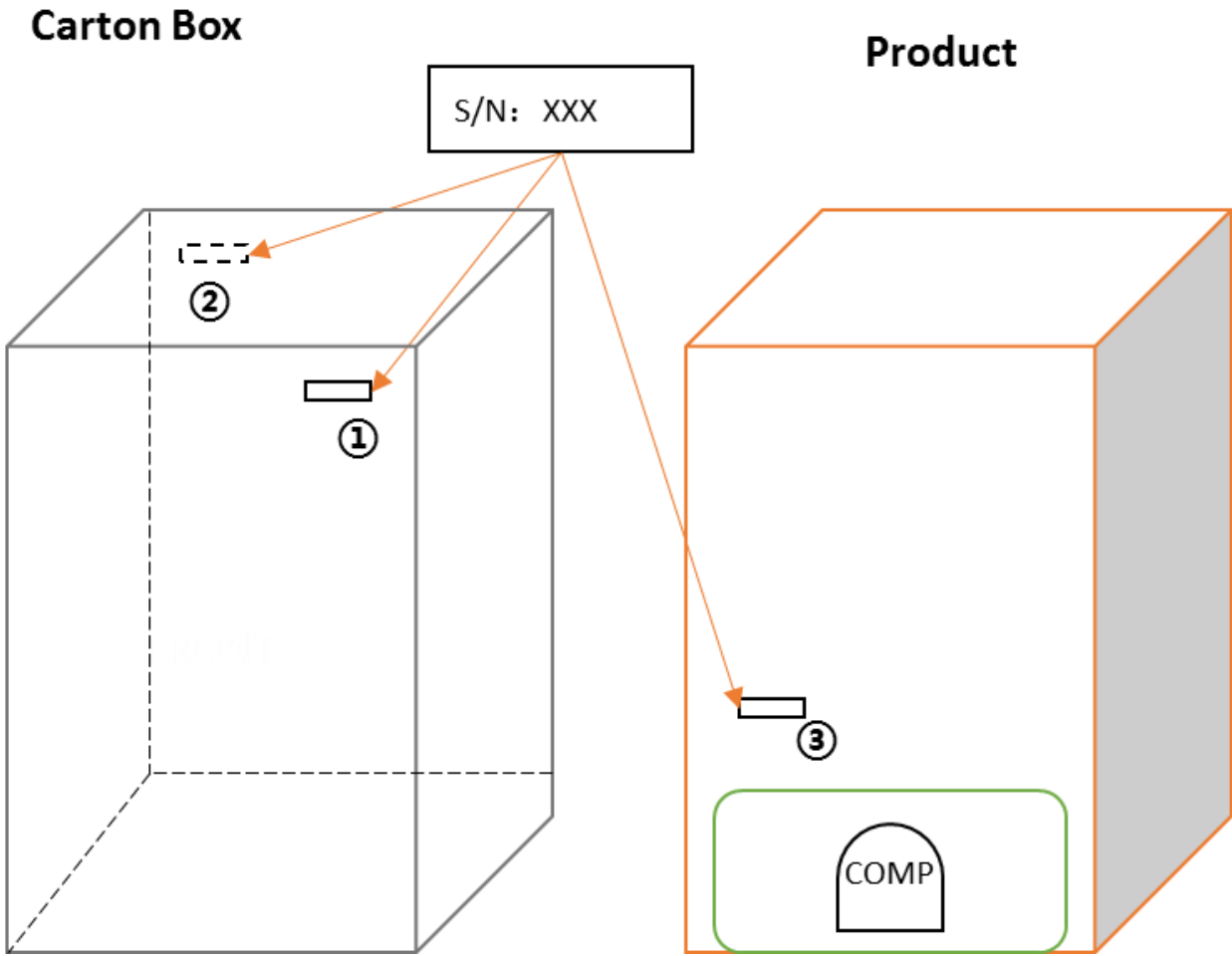
(The picture is only for reference)

4.3 Midea product serial number and location

1) **Product Serial Number** — Including order number, production date and other information. When the product occur problem, it needs to be recorded or photographed and provided to us.



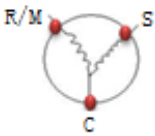
2) Paste location



Some products also have S/N on the lower part of the right side of the Cabinet.

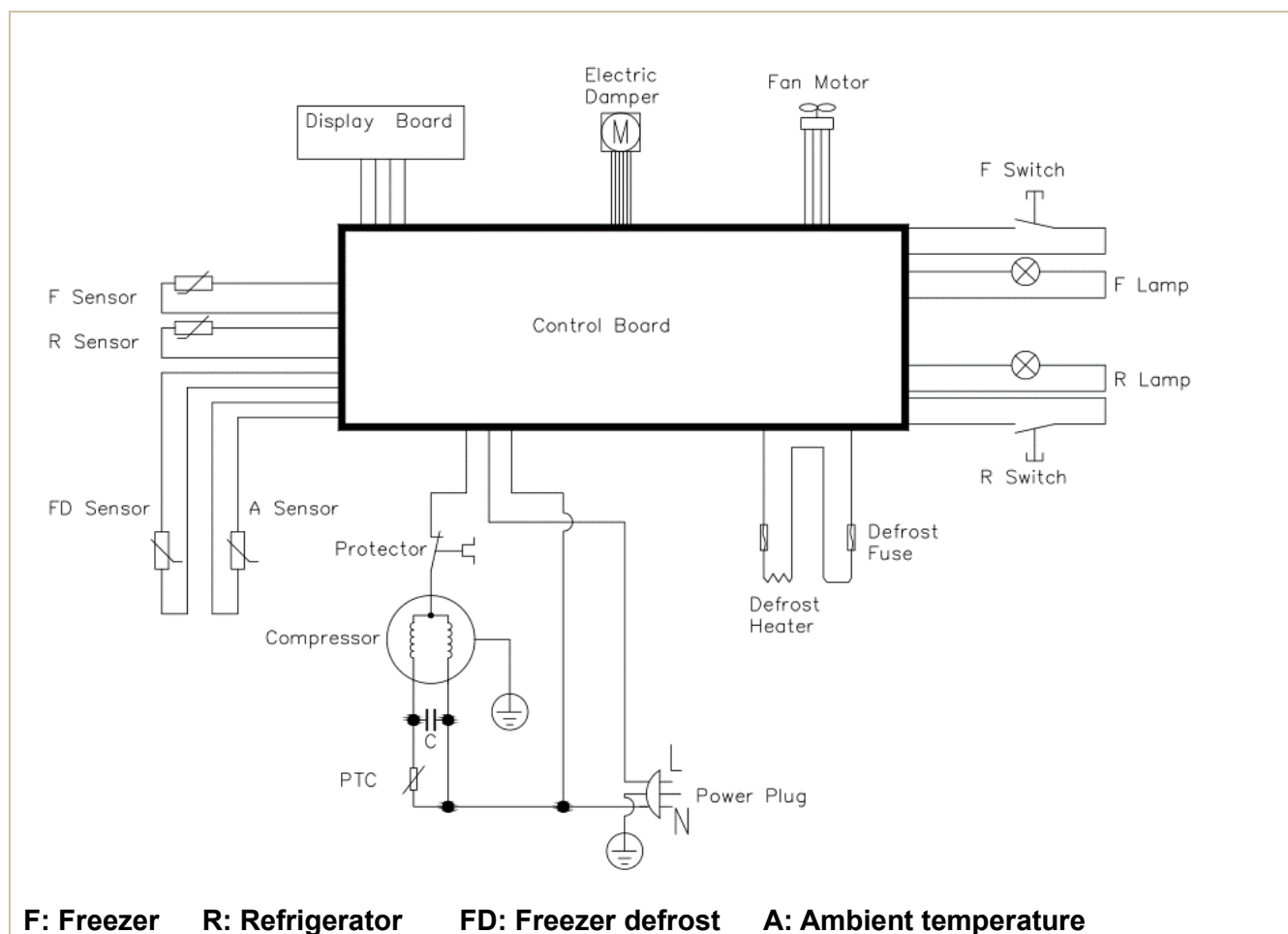
5. Electric control system

5.1 Electrical parts parameters

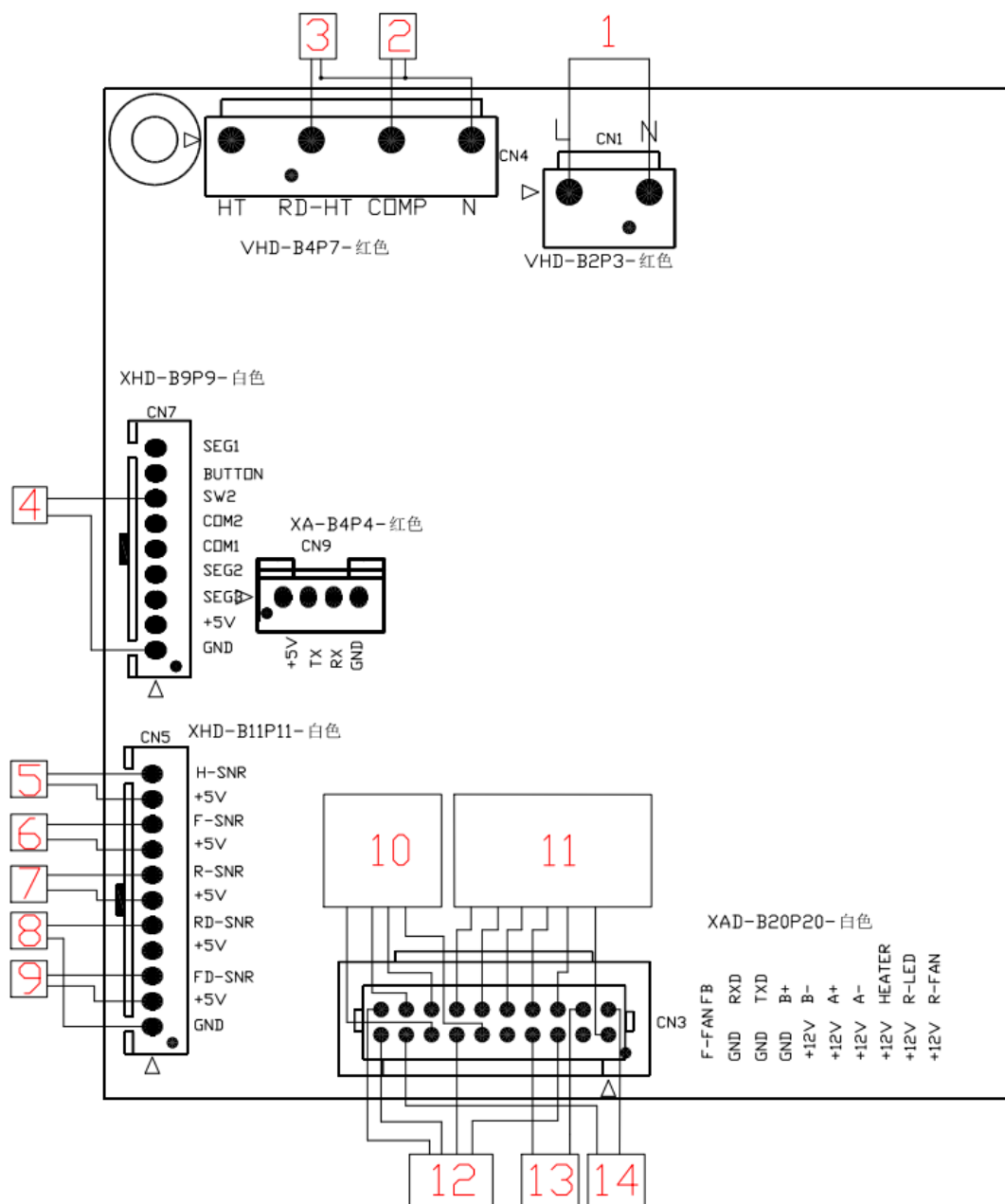
Applicable Model	HD-668FWEN	
Product Model	UR-BCD512WE-SQ	
Rated Voltage	115V、60Hz	
Item	Specification	Specification
Refrigerant	R600a	
Compressor	EZ65H1X-UQCL(MD#) (Part code : 11101010010152)	
Starting device type	Fixed Speed	
The COP of compressor	1.85(W/W)	
The max cooling capacity of compressor	133 W	
Winding resistance of compressor wiring terminal (20°C)	Rmc:7.07Ω±7% Rsc:6.07Ω±7% Rms = Rmc + Rsc	
Winding resistance picture		
Starter(PTC)	QPE2-A4R7MD3 V (Part co: 12031000001584)	
Overload protector(OLP)	DRB26T61A1 V (Part code : 12031000001301)	
Integrate PTC+OLP	None	
Variable frequency driver board	None	
Capacitor	10μF	
Power filter (EMI)	None	
Power reactor (EU EMC)	None	
Motor		
Fan motor of the freezing chamber	DC12V、2W、2300r/min	
Fan motor of the refrigerating chamber	None	
Electric damper	Have	
Lights		
Lights inside the refrigerating chamber	DC12V、2W	
Lights inside the freezing chamber	DC12V、2W	
Others Lights	None	
Switch of the light	■Mechanical switch	

	□Magnetism control switch	
Defrosting parts		
Defrosting sensor	NTC B3839 (B5/25=3839K±2%)	
Fuse in freezing chamber	AC115V、77(0 ~ -5) °C	
Defrost heater in freezing chamber	AC115V、215W	

5.2 Circuit diagram



5.3 Main PCB terminal connection diagram

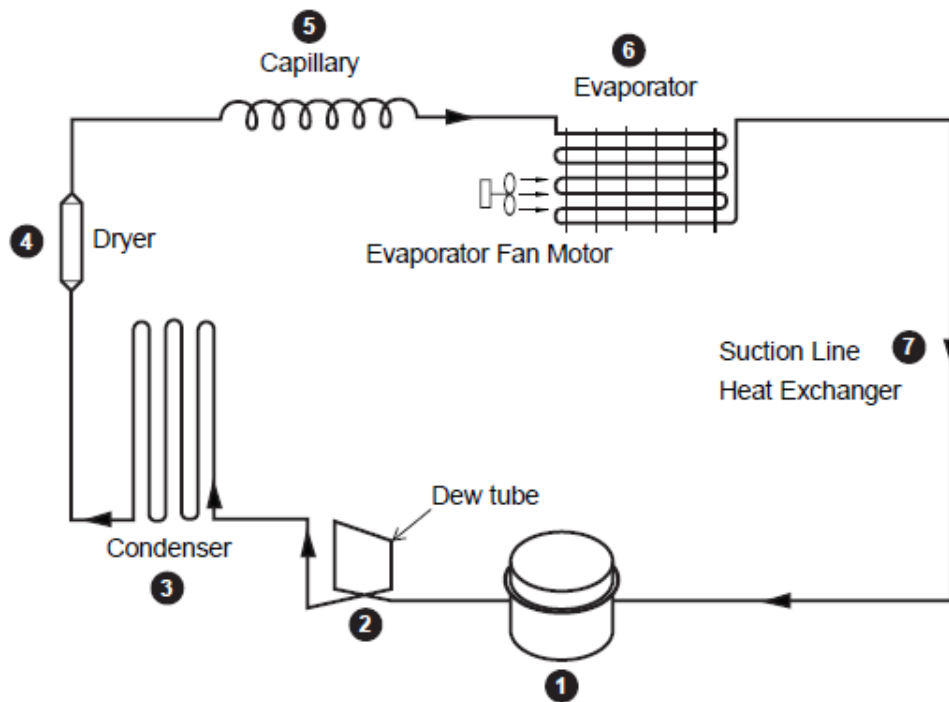


Connecting terminals	Connecting terminals
1. Power supply	8. Freezing light switch
2. Compressor	9. Freezing defrost sensor
3. Freezing defrost heater	10. Display control panel
4. Refrigeration light switch	11. Electric damper
5. Ambient temperature sensor	12. Fan of freezing chamber
6. Freezing temperature sensor	13. LED of freezing chamber
7. Refrigeration temperature sensor	14. LED of refrigeration chamber

6. Refrigeration system

6.1 Refrigeration system working principle

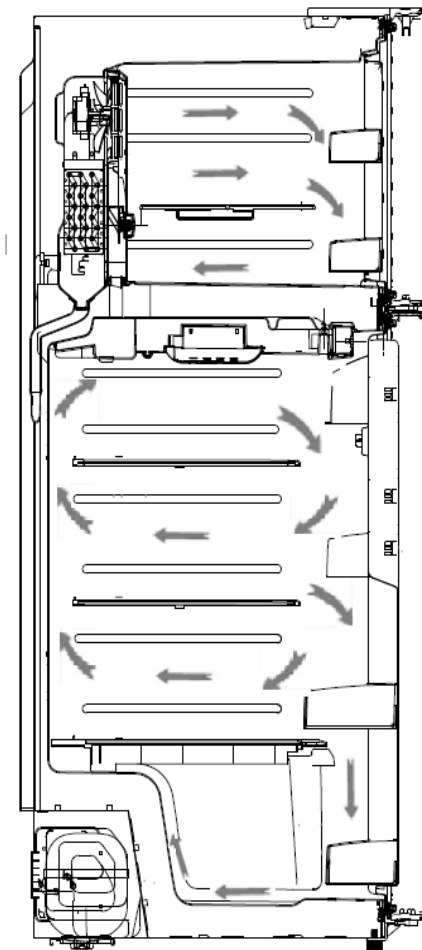
① Compressor → ② Anti-condensation tube → ③ Condenser → ④ Dry filter → ⑤ Capillary tube → ⑥ Evaporator → ⑦ Suction tube → ① Compressor



6.2 Cooling pipeline and drain pipe inside the cabinet

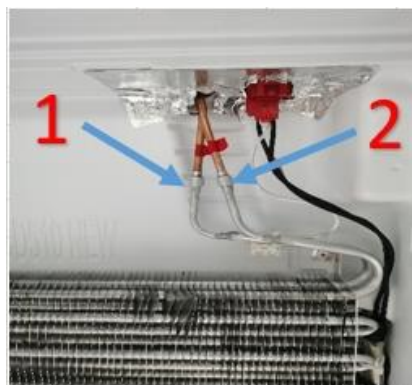


6.3 Circulating route of cooling air



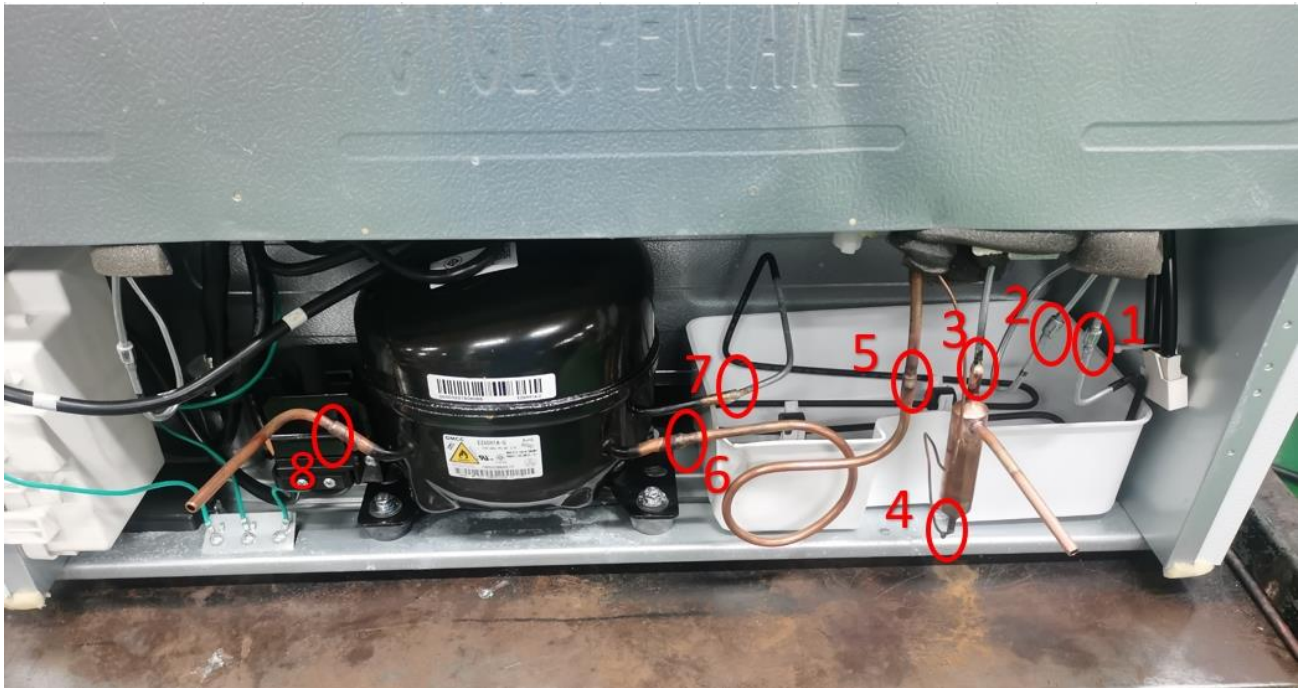
6.4 Welding points in chambers or foam layer

1) Welding points on freezer evaporator



Welding point	Pipe outer diameter (mm)	
1-Freezer capillary and inlet of evaporator	Copper pipe: $\Phi 6$	Aluminum pipe: $\Phi 6.0$
2-Heat transition tube and outlet of evaporator	Copper pipe: $\Phi 6$	Aluminum pipe: $\Phi 6.0$

6.5 Welding point in the compressor case



Welding point	Pipe outer diameter (mm)	
1-Inlet of anti-condensation tube and outlet of venting connection tube	Steel pipe: $\Phi 4.0$	Steel pipe: $\Phi 4.0$
2-Inlet of condenser tube and outlet of anti-condensation tube	Steel pipe: $\Phi 4.0$	Steel pipe: $\Phi 4.0$
3-Outlet of left condenser tube and inlet of dry filter	Steel pipe: $\Phi 4.0$	Copper pipe: $\Phi 5.0$
4-Outlet of dry filter and inlet of freezer capillary	Copper pipe: $\Phi 2.8$	Copper pipe: $\Phi 1.8$
5-Heat transition tube and Suction connection tube	Copper pipe: $\Phi 6.0$	Copper pipe: $\Phi 6.0$
6-Suction connection pipe and compressor intake tube	Copper pipe: $\Phi 6.0$	Copper pipe: $\Phi 8.16$
7-Compressor outlet tube and inlet of venting connection tube	Copper pipe: $\Phi 6.16$	Steel pipe: $\Phi 4.0$
8-Compressor process tube and refrigerant filling tube	Copper pipe: $\Phi 8.17$	Copper pipe: $\Phi 6.0$

7. Dismantling of parts

7.1 Parts on the door

Door seal

Door seal is installed into door liner groove.

- 1) Open the refrigerator door;
- 2) Take the door seal out of door liner;



Door tray

While squeezing it inward, lift up the door tray and take it out from door liner.



Door stopper

After removing the door, remove the fixing screws at the position shown in the figure, remove the door sleeve pipe and stopper, and complete the stopper replacement.



7.2 Parts inside the refrigerator

Refrigerator Fruit box cover

Remove the crisper cover of ref. compartement according to below steps:

- 1) Take out the crisper firstly.
- 2) Lift up the Fruit box cover, and then pull out it.



Shelves

Lift up the division plate with a proper force and pull it out towards yourself;



Freezer Drawer

None

None

7.3 Light system

Light

Light of the refrigerating

- 1) Remove the lamp cover



- 2) Remove the light.
- 3) The reverse process can complete installation.



Light of the freezing	
1) Remove the lamp cover	
2) Remove the light. 3) The reverse process can complete installation.	
There is a light switch on the side wall of the cold room and the cold room.1) Loosen the hook with small normal screwdriver and pull out the switch until the wire connector reveals.	 

7.4 Air duct components refrigerating chamber and fan motor

Air duct components in freezing chamber

- 1) The refrigerator air duct components is located on the back of the refrigerating chamber.
- 2) use slotted screwdriver to dismantl the deodorizer box. then use cross screwdriver to remove the fittin screw of air duct cover



- 3) use fingers to hold the button edge of air duct cover and pull it outwards with big strength



Fan motor of air duct

None

7.5 Air duct components in freezing chamber and fan motor

Air duct components in freezing chamber

- All accessories in the freezing chamber should be dismantled before removing the air duct components.
- 1) Remove 2 screws on the cover plate of the freezing air duct using a cross screwdriver



- 2) Hold the middle position upper of freezer air duct, raise it then pull it outwards gently.



3) Pull out the connector terminal.



4) Take out the freezing air duct assembly.



Fan motor of air duct

Pull buckles at the back of the freezing air duct and remove 2 front cover plates;



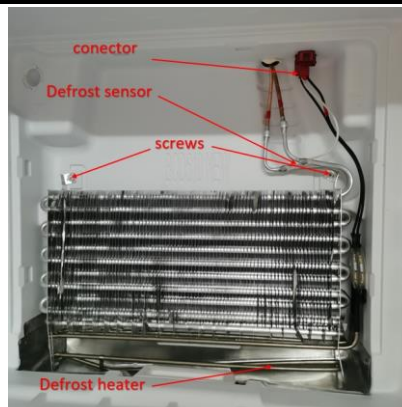
Replace the fan in reverse steps;



7.6 Evaporator and Defrost system

Evaporator in freezing chamber

- 1) Remove the air duct components in freezing and variable temperature chamber.
- 2) Disconnect all connecting terminals.
- 3) Remove the welding on inlet and outlet tubes.
- 4) Remove the evaporator.



Components on the evaporator

Defrost heater with defrost sensor and fuse, it can be replaced separately.

- 1) Cut off wiring terminal
- 2) Cut off the band which fixes the sensor
- 3) Cut off the band which fixes the fuse
- 4) Separate the sensor and the evaporator
- 5) Take off the defrost heater from the supporting plate of evaporator

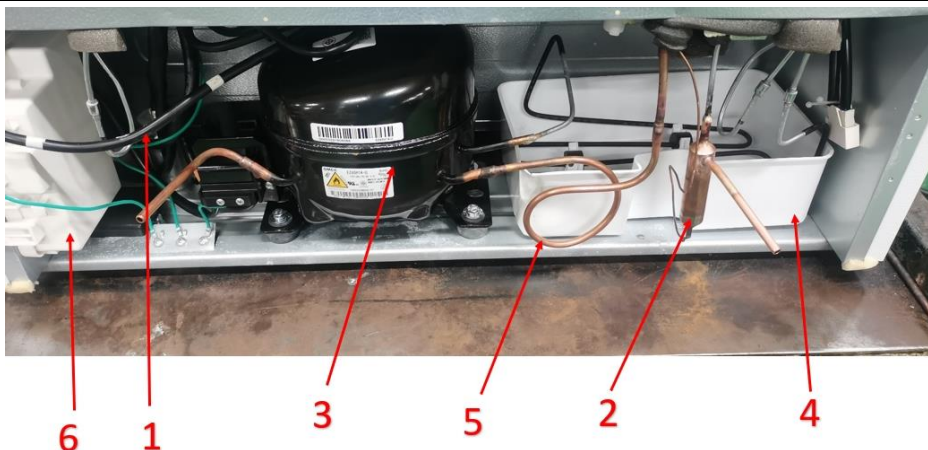


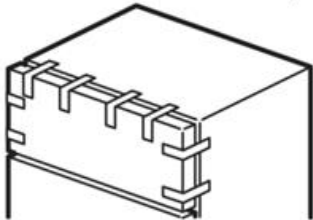
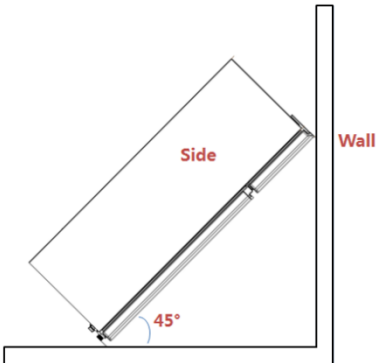
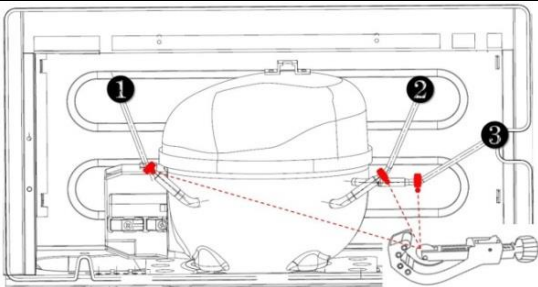
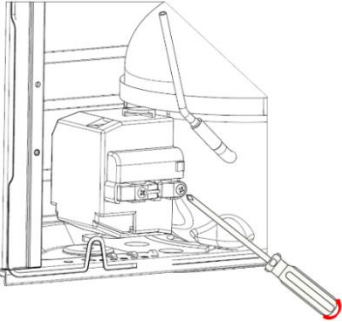
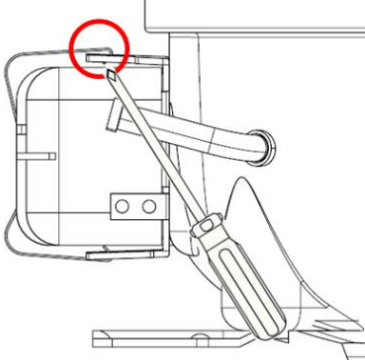
7.7 Compressor case

Rear cover

None

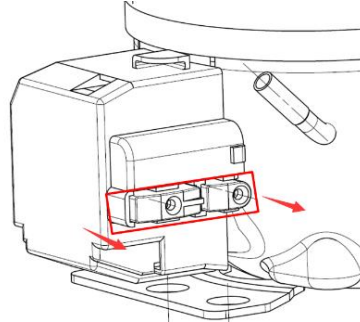
Piping and parts in the compressor case



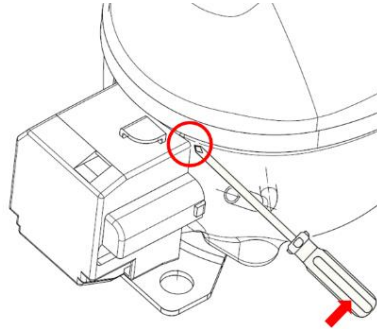
1-Power cord 2-Dry filter 3-Compressor	4-Drain tray 1 5-Suction connection tube 6-Drain tray 2
Disassembly and assembly of compressor	
1) Cut off the power, remove the goods in the refrigerator, with the tape to make the door fixed firmly and prevent the door dropping when the refrigerator dumping.	
2) Slowly tilt the refrigerator forward, relying on the wall or a solid enough object, leaving space to facilitate the operation. For safety, it should be carried by someone to prevent its falling.	
3) Cut off the compressor pipeline.-①Cut off the process pipeline.-②Cut off the low-pressure muffler.-③Cut off the high-pressure exhaust pipe.	
4-1) Remove the screws(for some models) -Two screws outside -One screw inside	
4-2) Remove the metal clamp(for some models) -Disassembly the metal clamp that is fix the electric appliance shield	

5) Remove the clipping strip

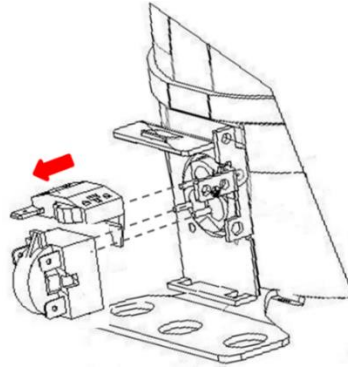
Slowly pull it out

**6) Remove the protective cover**

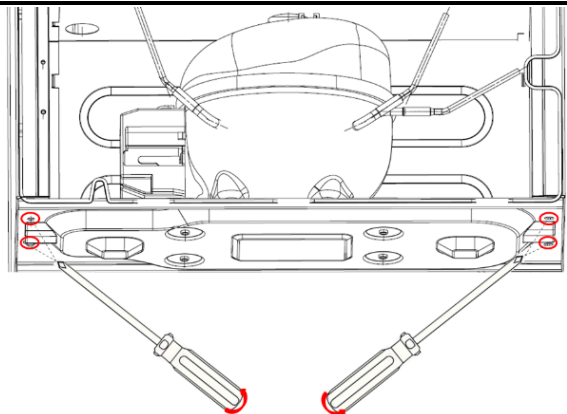
-Pry the protective cover slowly from the upper part,
-Pull it out and remove it.

**7) Remove the starter and protector**

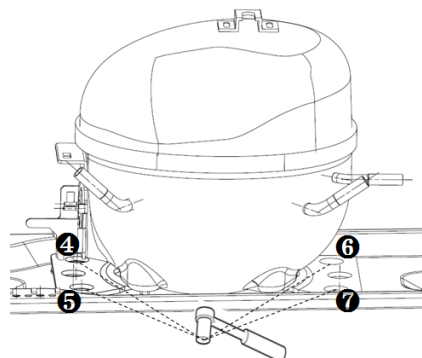
Unplug the starter and protector (you can use a screwdriver to pry it slowly)

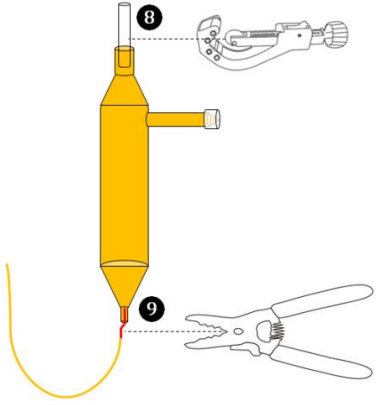
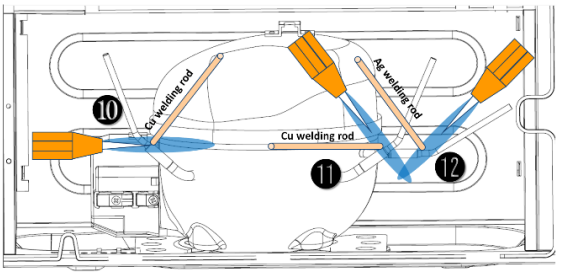
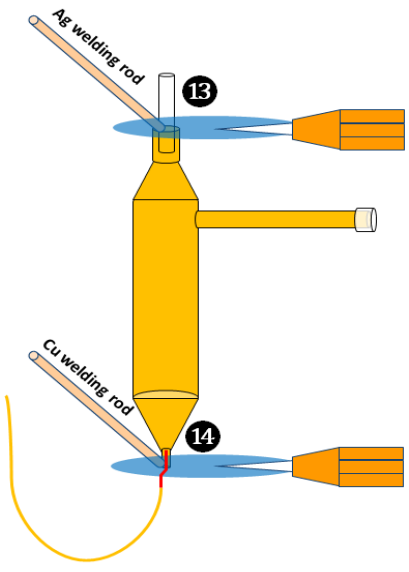
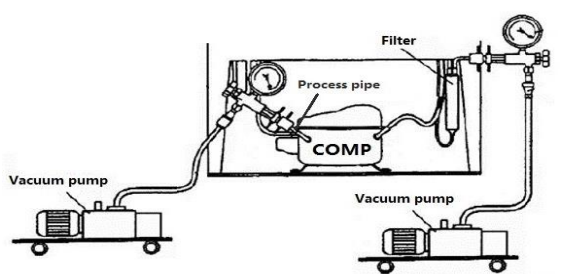
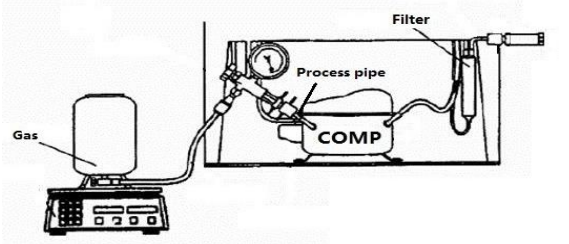


8) Loosen the screw of the compressor bottom plate, remove the floor together with the compressor from the box.

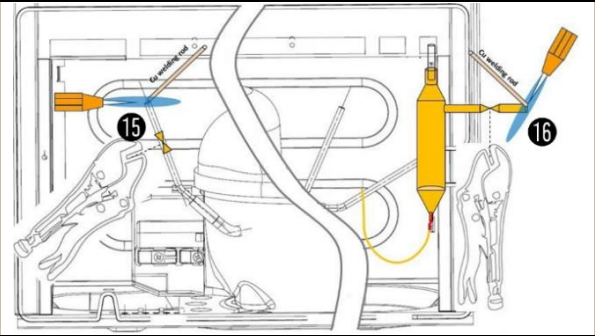


9) Use the wrench to remove the bolts by steps ④ ⑤ ⑥ ⑦, replace the compressor and reverse process can complete installation.



10) Use Pipe cutter cut off the condenser tube ⑧, then Shear off capillary ⑨ by the capillary tube scissors.	
11) Replace the compressor and welding the compressor pipeline.-⑩Welding the process pipeline.-⑪Welding the low-pressure muffler.-⑫Welding the high-pressure exhaust pipe.	
12) Replace the filter, Cu-Fe tubes welding ⑬ used Ag welding rod, Cu-Cu tubes welding ⑭ used Cu welding rod.	
13) Vacuum system, The degree of vacuum below 6Pa.	
14) Perfusion refrigerant.	

15) Use the vise grip pliers clamp the middle of the process pipe, then seal welding process tube 15 16.



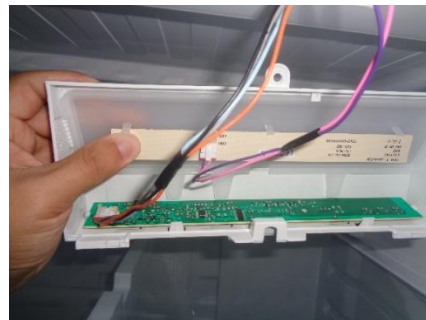
7.8 Display control board

Display control board

1) Remove one screw using a Phillips screwdriver

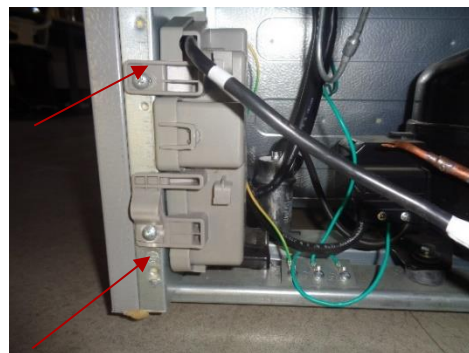


2) Remove the terminal and the display board.



7.9 Main control board

1) Remove the two screws that secure the main control board box.

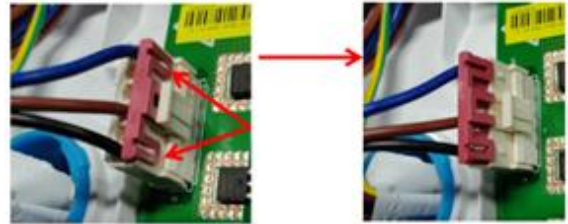


2) Take out the mounting box, pry up the cover, disconnect the wiring terminal.

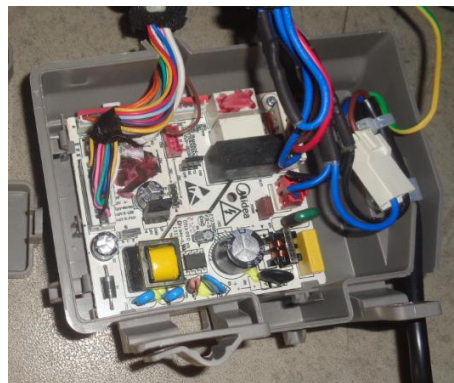


The connecting terminals remove:

- a. Use a needle tool to remove the lock at the arrow and remove the lock upward
- b. After the lock is removed, press the hook to remove the wiring connectors



5) Disconnect the wiring terminal, take out the main control board, and complete the replacement.



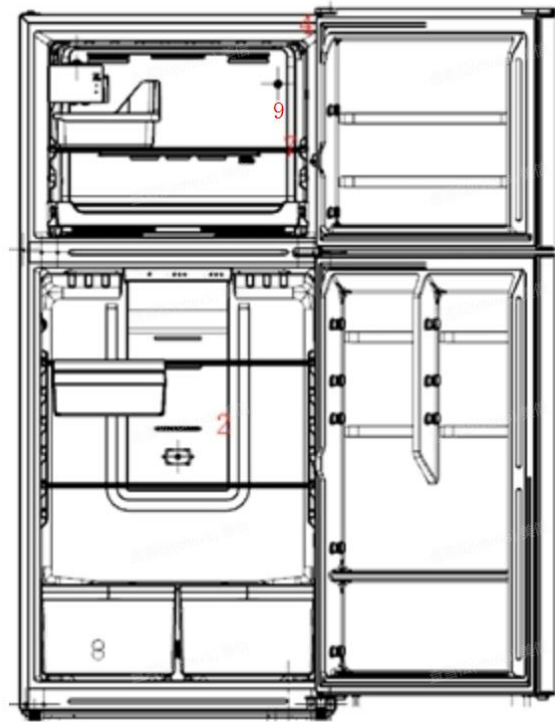
8. Temperature sensing system

8.1 Position of sensors

Position of sensors

Have 3 sensors

- ② Sensor in refrigerating chamber
- ④ Ambient temperature sensor
- ⑦ Defrost sensor in freezing chamber
- ⑨ Freezer sensor



8.2 Replacement of sensors

Sensor in refrigerating chamber

The refrigerating air duct components should be dismantled before removing this sensor.

- 1) Strip the tape that secures the wiring harness.
- 2) Then remove the sensor.



Ambient temperature sensor

The sensor is located in the hinge cover of the refrigerator top.



Defrost sensor in freezing chamber

- 1) Cut off wiring terminal;
- 2) Cut off the band which fixes the sensor;

**Freezer sensor**

- 1) Remove the air duct assembly of freezer (two screws);
- 2) Remove the air duct front plate;
- 3) Then remove the sensor.

**8.3 Sensor without terminal replacement****Sensor replacement guidelines**

Cut off the damaged head of sensor.

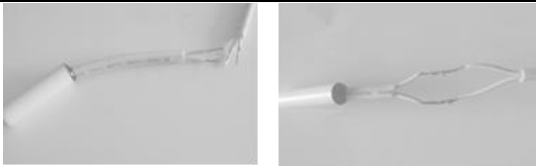


Strip off the sensor wiring.



Take out a new sensor to cut the head of sensor.
(Spare parts code: 11201007000795) Its technical specifications apply to all MIDEA refrigerators.



Strip off the head of the sensor and connect it.	
Wrap the two wires together with insulation tape.	
Wrap the two wires together.	

8.4 Sensor R/T table

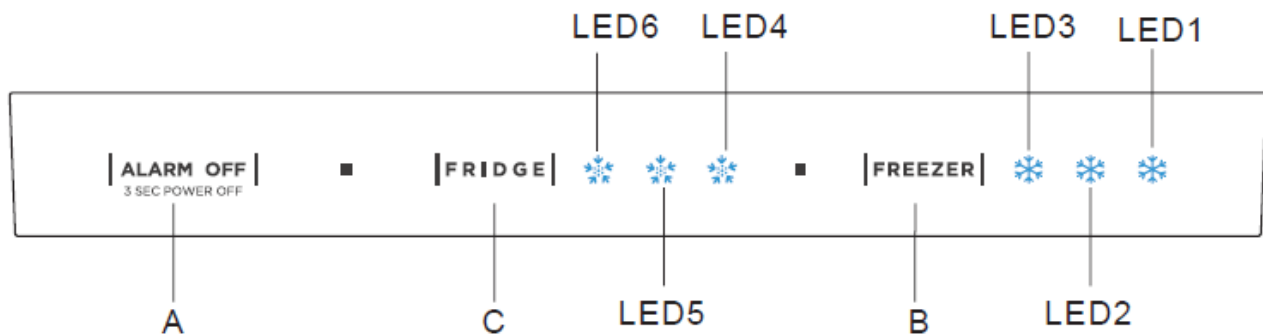
T _x (°C)	T _x (°F)	R (KΩ)	T _x (°C)	T _x (°F)	R (KΩ)	T _x (°C)	T _x (°F)	R (KΩ)
-30	-22.00	33.81	-5	23.00	8.392	20	68.00	2.501
-29	-20.20	31.85	-4	24.80	7.968	21	69.80	2.391
-28	-18.40	30.01	-3	26.60	7.568	22	71.60	2.287
-27	-16.60	28.29	-2	28.40	7.190	23	73.40	2.188
-26	-14.80	26.68	-1	30.20	6.833	24	75.20	2.094
-25	-13.00	25.17	0	32.00	6.495	25	77.00	2.005
-24	-11.20	23.76	1	33.80	6.175	26	78.80	1.919
-23	-9.40	22.43	2	35.60	5.873	27	80.60	1.838
-22	-7.60	21.18	3	37.40	5.587	28	82.40	1.761
-21	-5.80	20.01	4	39.20	5.315	29	84.20	1.687
-20	-4.00	18.90	5	41.00	5.060	30	86.00	1.617
-19	-2.20	17.87	6	42.80	4.818	31	87.80	1.550
-18	-0.40	16.90	7	44.60	4.589	32	89.60	1.486
-17	1.40	15.98	8	46.40	4.372	33	91.40	1.426
-16	3.20	15.12	9	48.20	4.167	34	93.20	1.368
-15	5.00	14.310	10	50.00	3.972	35	95.00	1.312
-14	6.80	13.550	11	51.80	3.788	36	96.80	1.259
-13	8.60	12.830	12	53.60	3.613	37	98.60	1.209
-12	10.40	12.160	13	55.40	3.447	38	100.40	1.161
-11	12.20	11.520	14	57.20	3.290	39	102.20	1.115
-10	14.00	10.920	15	59.00	3.141	40	104.00	1.071
-9	15.80	10.350	16	60.80	2.999	41	105.80	1.029
-8	17.60	9.820	17	62.60	2.865	42	107.60	0.989
-7	19.40	9.316	18	64.40	2.737	43	109.40	0.951

-6	21.20	8.841	19	66.20	2.616	44	111.20	0.914
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9. Function and operation

9.1 Display operation panel

Refrigeration chamber temperature control panel	
Button	Icons
A. Freezing chamber temperature gear setting	① Gear 1 (LED3) ② Gear 2 (LED2/3) ③ Gear 3 (LED1/2/3)
B. Refrigerating chamber temperature gear setting	① Gear 1 (LED6) ② Gear 2 (LED5/6) ③ Gear 3 (LED4/5/6)
C. ALARM set OFF	① Cancel alarm tone ② Standby mode (Fridge stop working)



9.2 Temperature control

9.2.1 Temperature setting of refrigerating chamber

Press the gear setting button 'SET' once, the gear will be changed once, 5 seconds after setting the gear, the refrigerator will be running in accordance with the set value.

Temperature gear of refrigerator chamber	Gear 1	Gear 2	Gear 3
Temperature value	8°C	4°C	2°C

9.2.2 Temperature setting of freezing chamber

Temperature gear of freezer chamber	Gear 1	Gear 2	Gear 3
Temperature value	-16°C	-19°C	-24°C

9.3 Mode setting

9.3.1 Standby mode setting

- 1) When the refrigerator is working, keep pressing the “” button for 3 seconds, the refrigerator will stop working.
- 2) When the refrigerator is in standby mode, keep pressing the “” button for 3 seconds, the refrigerator will operate normally.

9.3.2 Locking and unlocking (None)

9.4 Defrosting function

◆ Defrosting theory:

The defrosting of evaporator is realized by the heating of heater, following the temperature rise, the frost on evaporator becomes water, and the water flow into the evaporating pan via the draining system, the water in evaporating pan evaporate away finally

◆ Defrosting steps:

The fan motor stop work--- Compressor shutdown--- the heater start working ---the heater stop when the temperature rise to setting--- after 7 minutes, if the start-up condition is reached, the compressor will start--- after 3 minutes, the fan motor will start work

◆ Meet one of the following conditions , defrost heating exit:

- 1) When the frozen defrosting sensor has no fault, the measured temperature $T_{fd} \geq$ the set temperature which defrost heating exit
- 2) When the frozen defrosting sensor has no fault, the defrosting time is $\geq$ 60 minutes
- 3) When the frozen defrosting sensor is faulty , the heater will stop after work 20 minutes

9.5 Open door alarm (None)

9.6 Error code and solutions

Error code	Fault Type	Troubleshooting and Solutions
LED① and LED② together lighten	Refrigerator chamber sensor failure (E1)	Step 1: Check whether the connection terminals are plugged in place and whether there are foreign matters in them; after cleaning the terminals, plug them in again. Step 2: If the fault still occurs, pull out the corresponding connection terminal on the main PCB, use a multimeter to check the resistance value of the sensor, and confirm whether it is normal. Step 3: If the resistance value is wrong, replace the sensor. Step 4: If the fault still occurs, replace the main PCB.
LED① and LED③ together lighten	Freezer defrost sensor failure(E5)	
LED① and LED④ together lighten	Ambient temperature sensor failure (E7)	
LED② and LED④ together lighten	Communication failure (E6)	
LED① and LED⑤ together lighten	Freezing chamber sensor failure(E2)	

9.7 Test mode

Test items	Setting Method	Setting result
Enter Test Mode	Keep pressing the Freezing and Refrigerating chamber gear setting button for 3 seconds and release	
Select to enter into forced defrosting mode in all rooms	Press the "freeze" button to adjust and select the mode, as shown in the picture on the right; The lock takes effect.	
	In forced defrosting mode, when the freezing defrosting sensor reach a temperature of 8°C and the defrosting heater has been working for at least 3 minutes.	then the refrigerator will exit the test mode and return to normal operation mode
	In forced defrosting mode, if the defrosting heater has been working for 60 minutes,	then the refrigerator will exit the test mode and return to normal operation mode
Select to exit the test mode	In test mode, power off or enter standby mode	The refrigerator will exit the test mode
	Keep pressing the Freezing and Refrigerating chamber gear setting button for 3 seconds and release	
	After entering the test mode, press the "freeze" button to adjust the mode selection mode, as shown in the picture on the right; The lock takes effect.	
	Control rules for forced cooling mode: 1, the compressor is the most high-grade operation; 2, the highest grade operation of the refrigeration fan; 3, air door open;	
	Exit mode after 72 hours. In the forced mode, press the cold temperature adjustment button and the cold temperature adjustment button for 3s, lock and exit	

9.8 Demo mode (None)

9.9 Backup data for power fail

- 1) The running state of the refrigerator is remembered after compressor running for 1 hour continuously.
- 2) The running state of the refrigerator is remembered after change function settings and lock.
- 3) When the refrigerator is out of power and recharged, the running state of the refrigerator is same as before.

10. Compressor

10.1 Compressor on and off control specifications

1.1 When one of the following conditions is met, the compressor stops:

- 1) Measured temperature in Freezing chamber $\leq$ Shutdown temperature of Freezing chamber
- 2) Enter defrosting cycle
- 3) The compressor runs continuously for more than 5 hours, will stop at least 10 minutes

1.2 When compressor shutdown time has been more than 10 min, one of the following conditions is met, the compressor starts up:

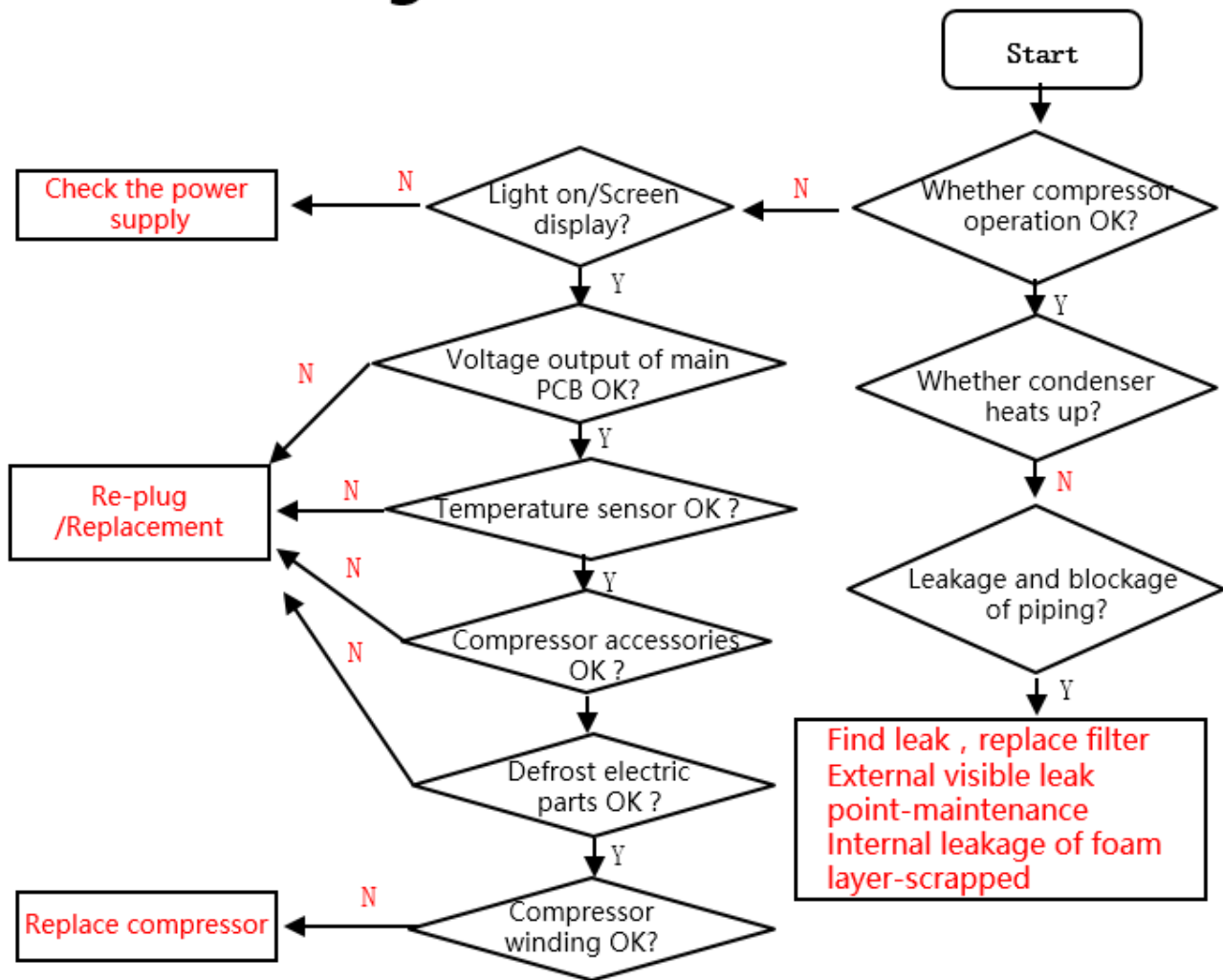
- 1) Measured temperature in Freezing chamber $\geq$ Start up temperature of Freezing
- 2) Refrigerator is set to forced cooling mode

★When 1.1 and 1.2 are not satisfied, the compressor maintains the original state

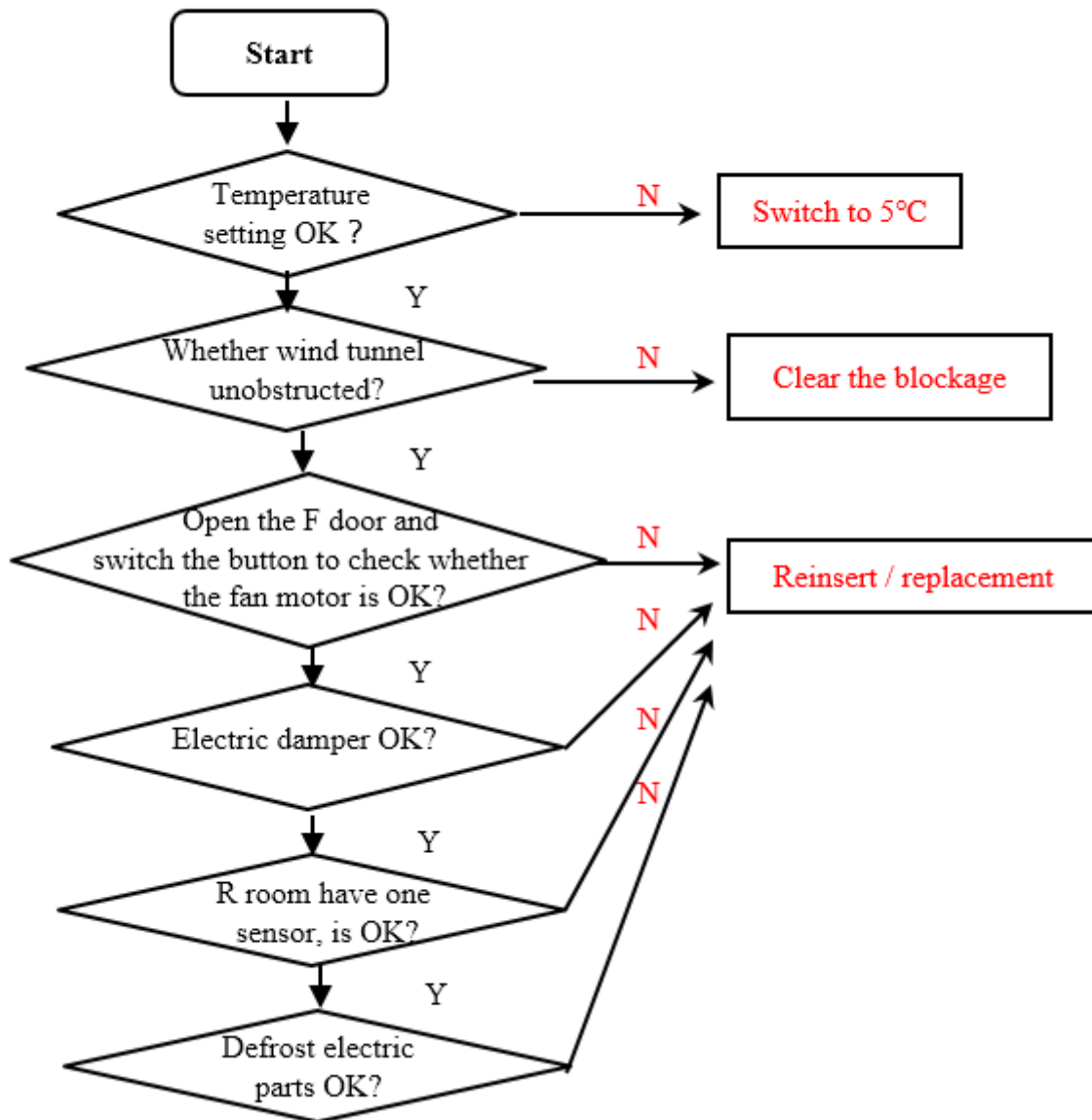
11. Troubleshooting Method

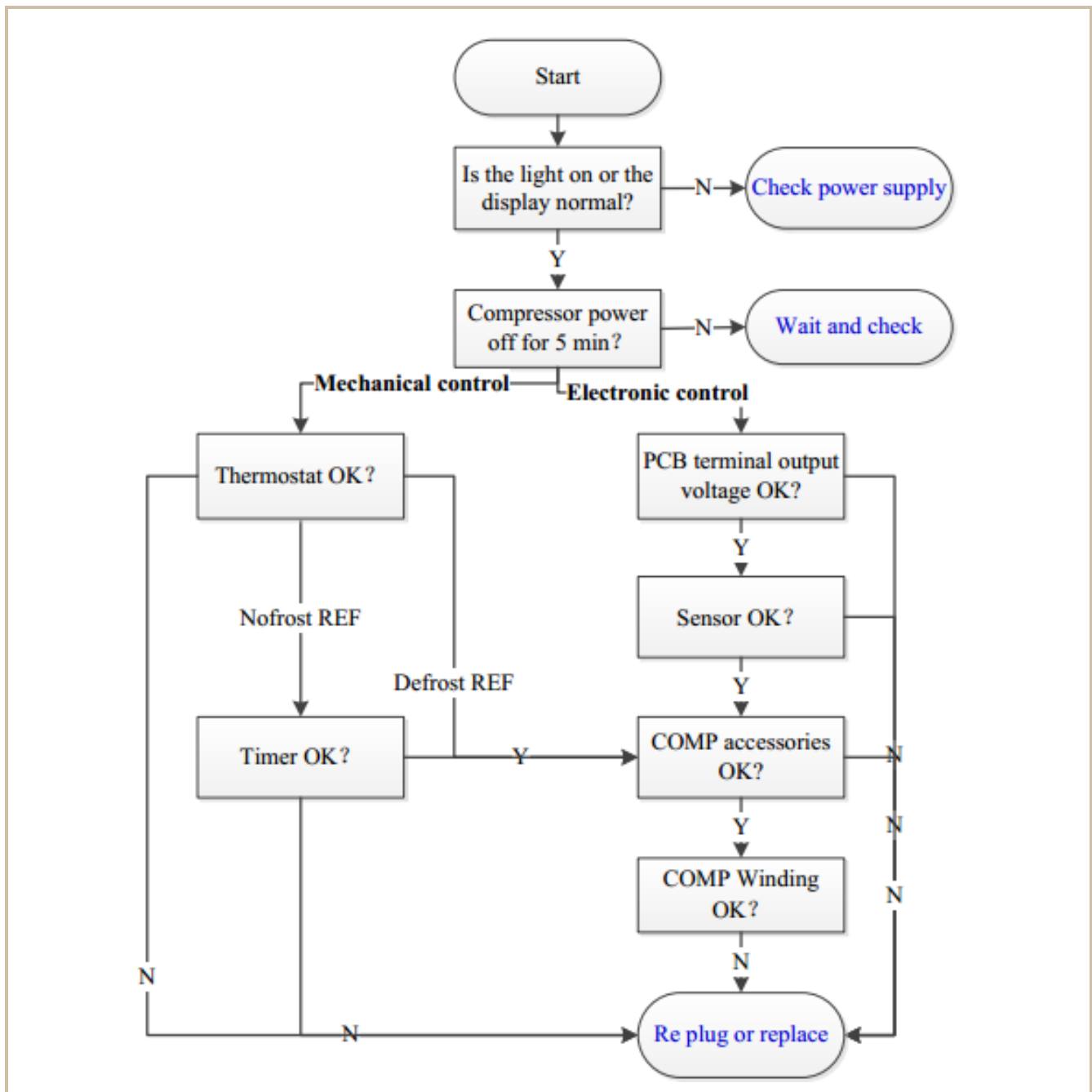
11.1 No cooling (Air cooling-Electronic)

No cooling of F room and R room



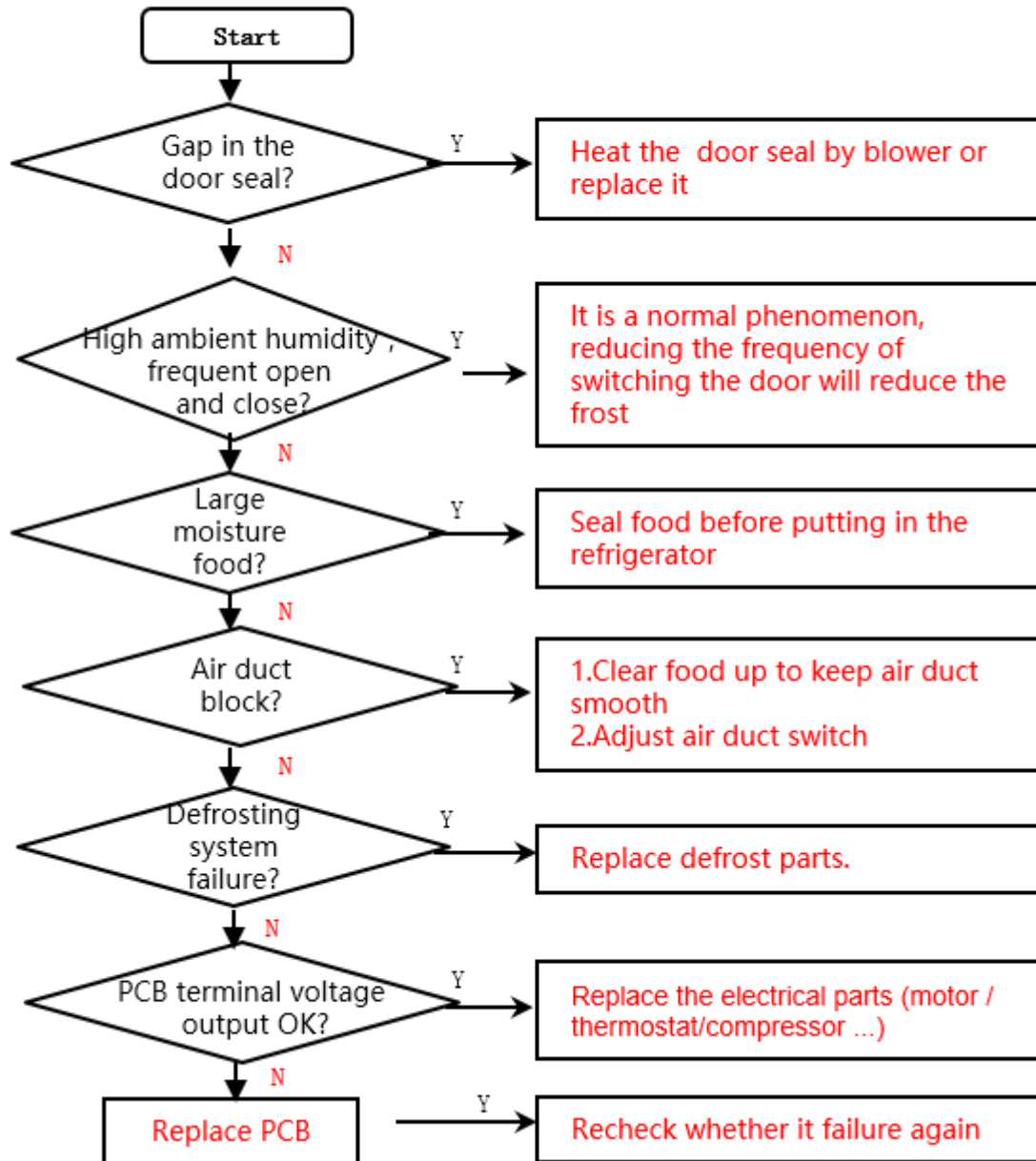
F room cooling, R room no cooling



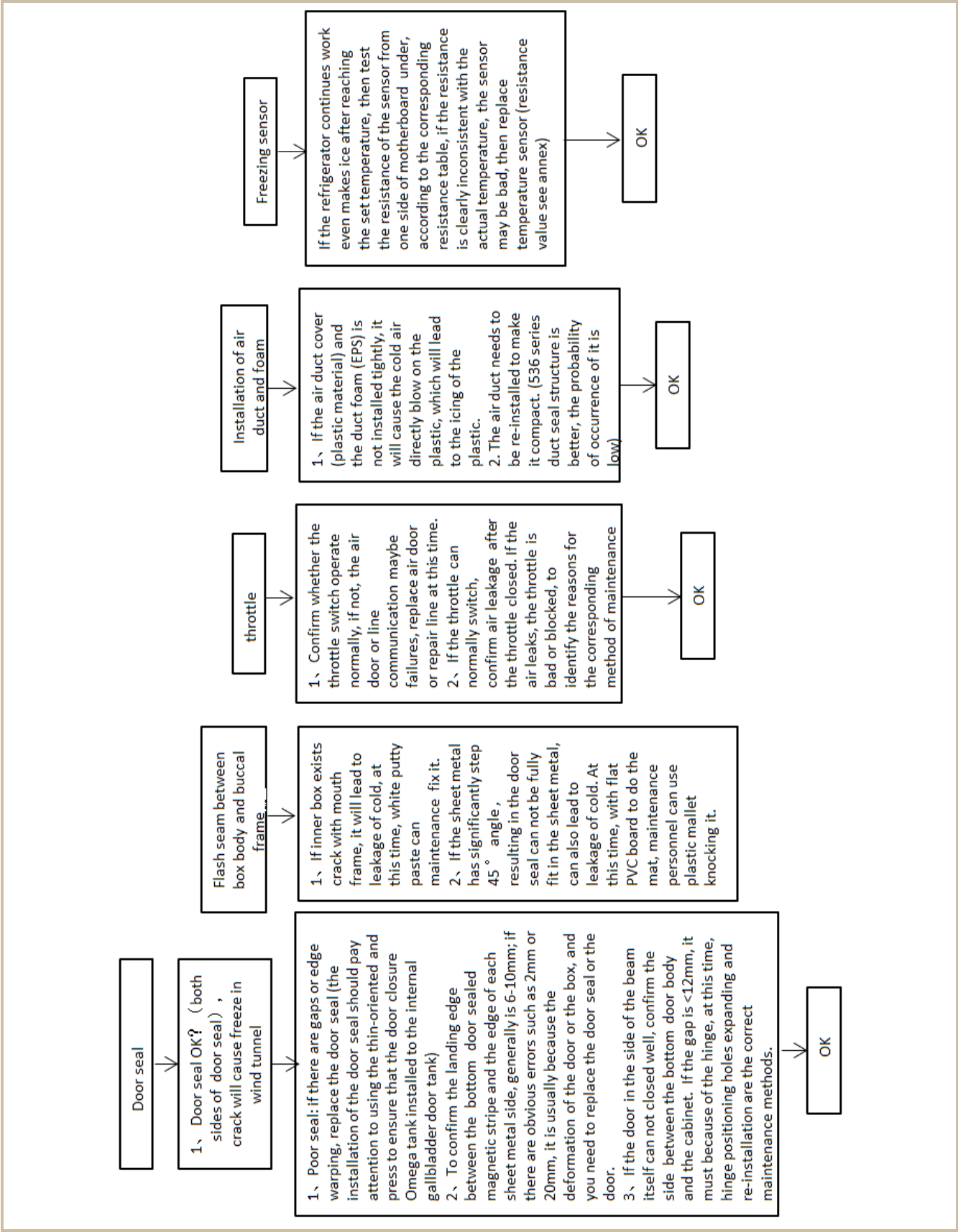
11.2 No working of compressor

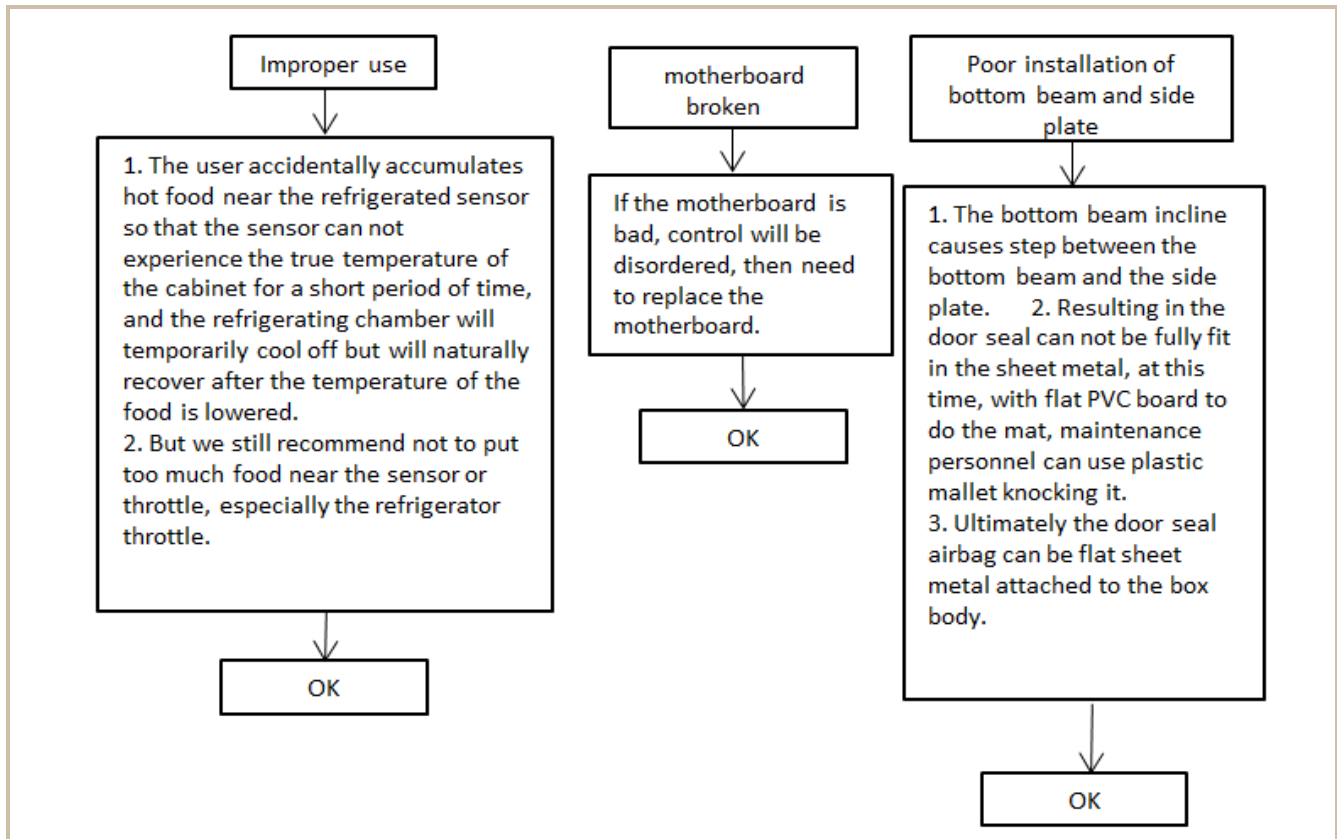
11.3 Inside frosting, no defrosting

Inside frosting, no defrosting

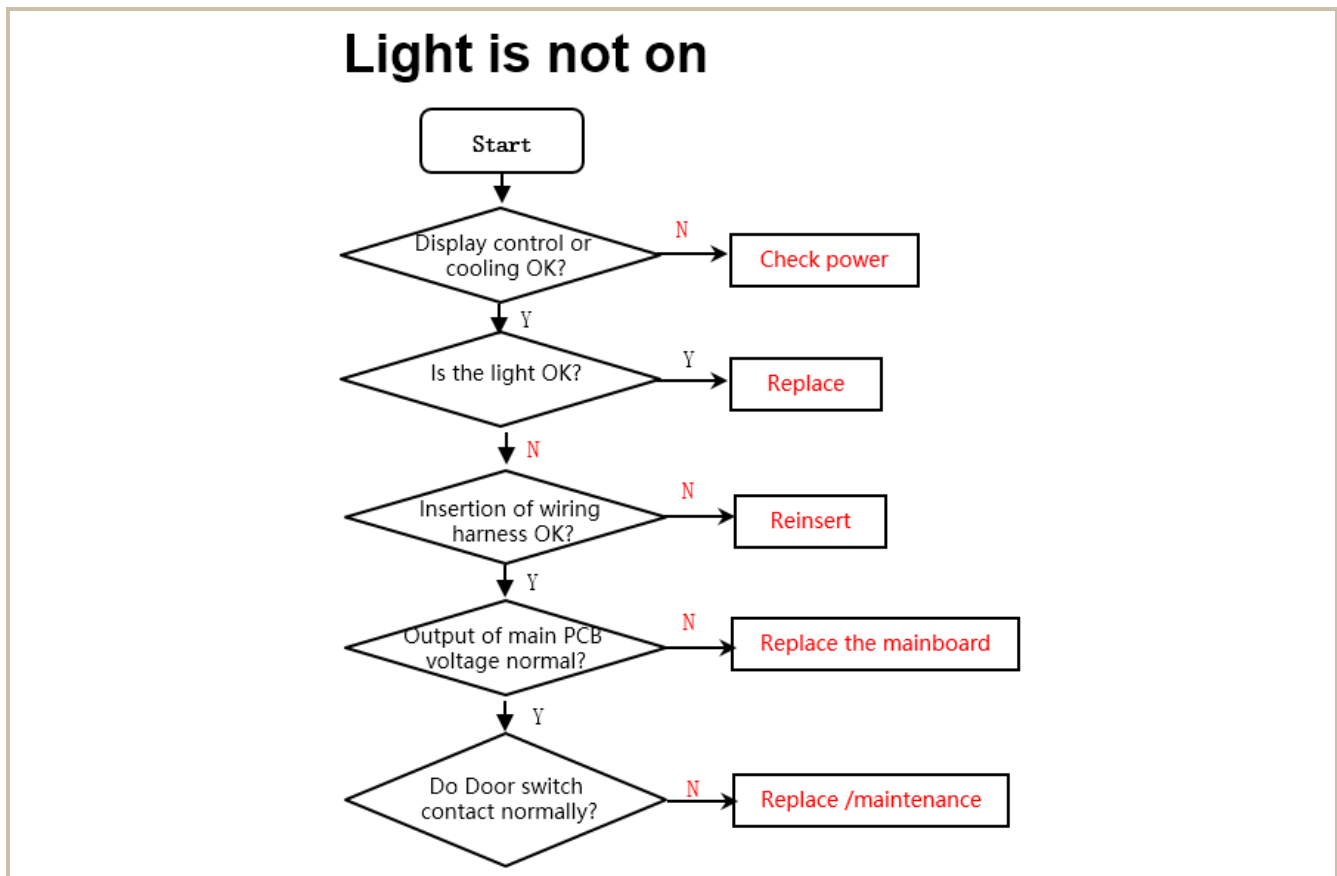


11.4 Inside frosting, no defrosting-Maintenance guidelines

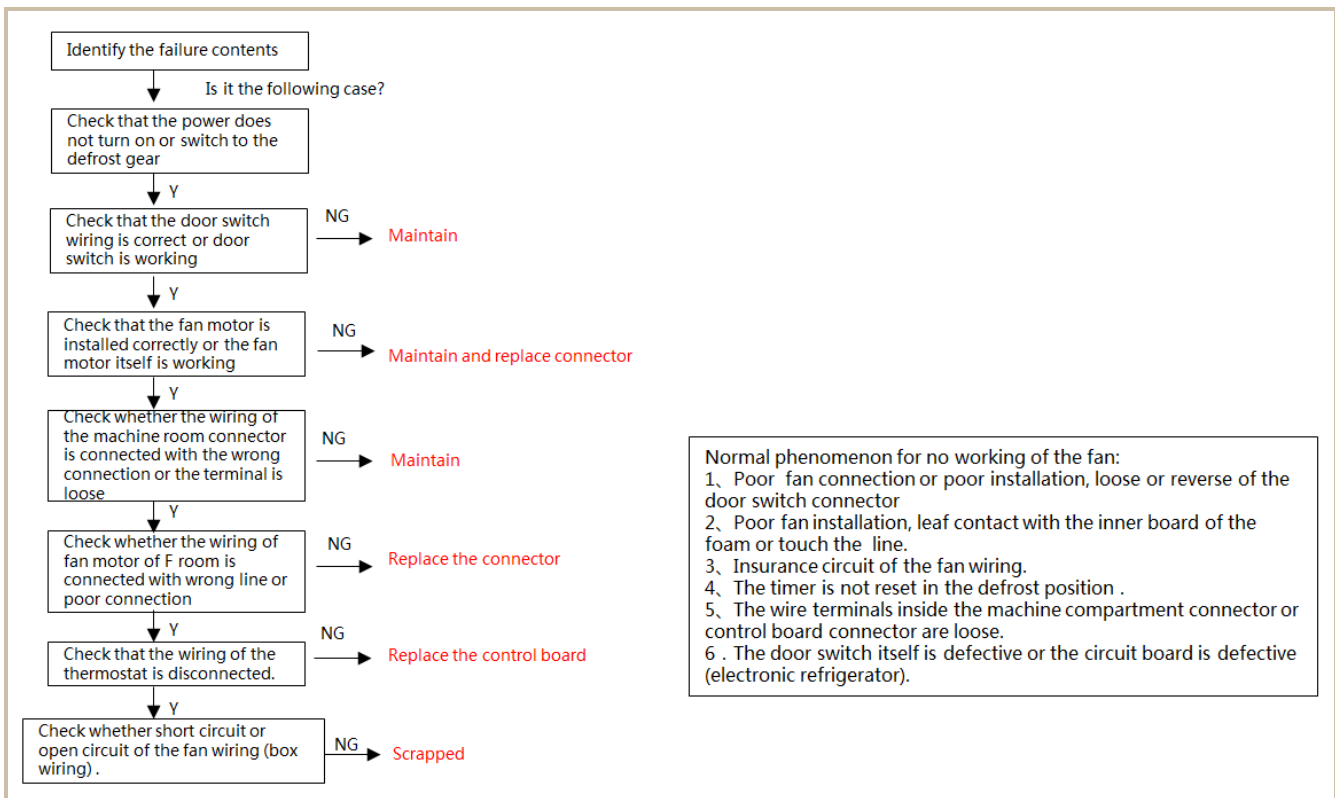




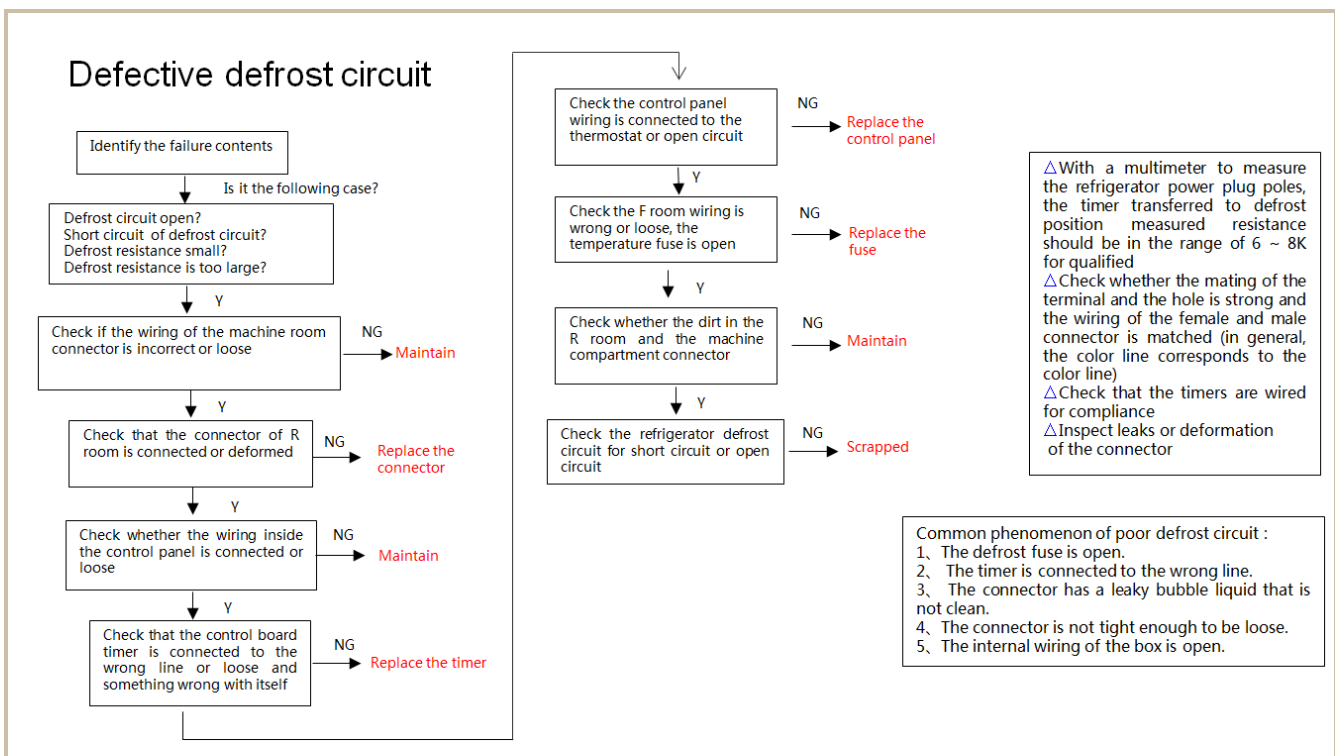
11.5 Light is not on



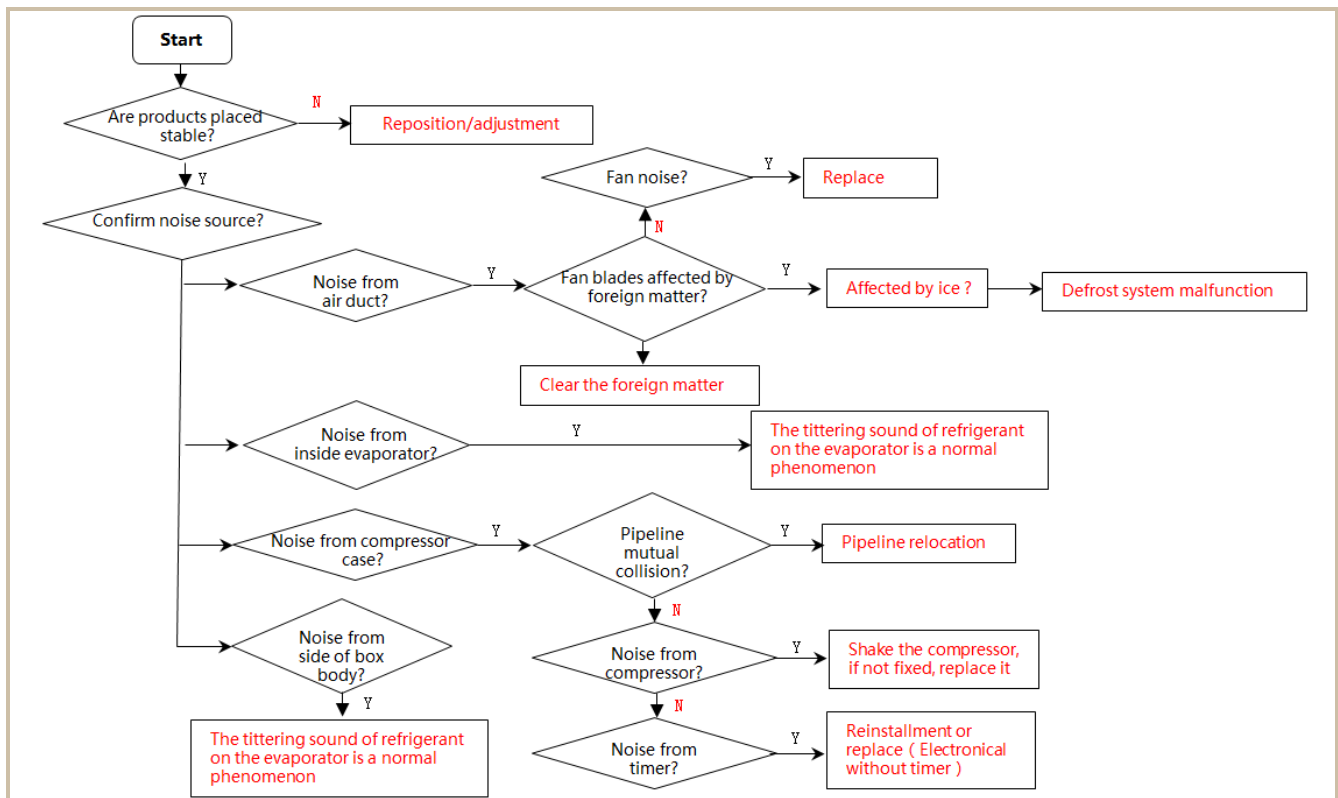
11.6 Fan failure



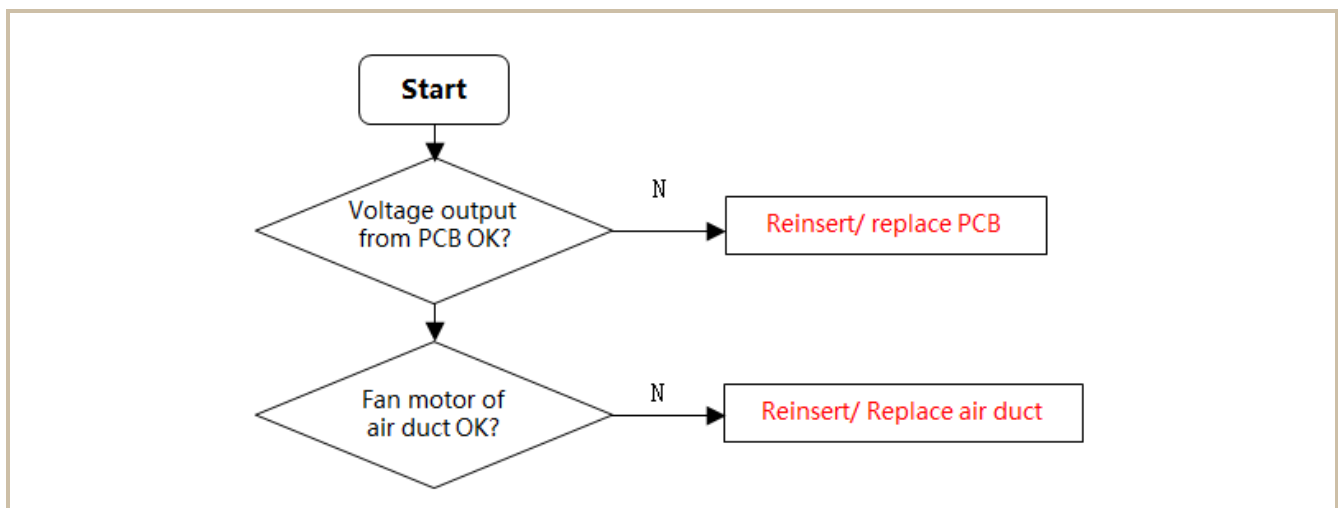
11.7 Defective defrost circuit



11.8 Noise



11.9 Air duct not operated



12. Product exploded view and spare parts list

Please log in to TSP system to view and download these contents.

Entry Guidelines (TSP System)

■ Where



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■ How

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Preliminary password: Please contact the TSP administrator or sales manager for the password.

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Administrator:

REF: Jamie Liu(liujc5@midea.com)



The symbol on the product or its packaging indicates that this product must not be disposed of with your other household waste. Instead, it is your responsibility to dispose of your waste equipment by handing it over to a designated collection point for the recycling of waste electrical and electronic equipment. The separate collection and recycling of your waste equipment at the time of disposal will help conserve natural resources and ensure that it is recycled in a manner that protects human health and the environment. For more information about where you can drop off your waste for recycling, please contact your local authority, or where you purchased your product.

Midea Refrigerators

If you need to get detailed technical information from the manufacturer, please contact:

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**Refrigeration Division
Overseas Sales Company**

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